Output, income and capital flows

How do real resources become income, saving and financial claims?

The seven-class route

Class	Question
1. Accounting and flows	Who produces, spends and finances?
2. Money, inflation and activity	How does policy affect demand?
3. Bonds and yield curves	How are future payments priced?
4. Foreign exchange	How are currencies priced?
5. Open-economy policy	Which adjustment is possible?
6. Crises and the euro	How does financial fragility spread?
7. Debt and macro trades	How do fiscal and market risks interact?

LECTURE 1

Production and income

Why can we measure the same economy in three ways?

A loaf of bread



Invented bakery economy. The farmer, mill and bakery each add value.

Value added along the chain

Producer	Sales	Bought inputs	Value added
Farmer	10	0	10
Mill	25	10	15
Bakery	60	25	35
Total			60

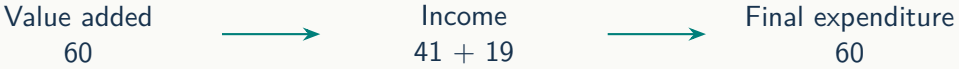
One period, one currency, no taxes or imports. Bread is the final good.

Output generates income

Producer	Wages	Operating surplus	Value added
Farmer	6	4	10
Mill	10	5	15
Bakery	25	10	35
Total	41	19	60

Each unit of value added accrues to labour or the business owners.

Three measurements of one flow



Production, income and expenditure count the same activity.

GDP counts current domestic production

Final goods and services produced inside the economy.

Investment includes new capital and changes in inventories.

Purchases of existing shares or old houses are asset trades.

Nominal output and real output

$$GDP_t^{\text{nominal}} = Q_t Y_t.$$

Y_t: real output. Q_t: the GDP price deflator.

Nominal growth combines more production and higher prices.

Growth rates separate prices and quantities

$$1 + g_{\text{nominal}} = (1 + g_{\text{real}})(1 + \pi_{\text{GDP}}).$$

$$1.03 \times 1.02 - 1 = 5.06\%.$$

Hypothetical 3% real growth and 2% GDP-deflator inflation.

Reading the GDP calendar

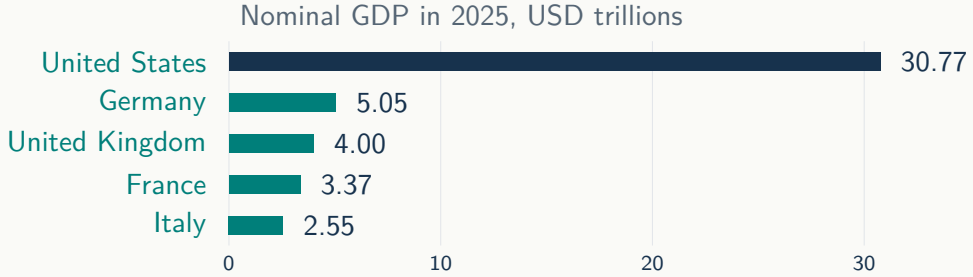
The US and euro-area countries measure GDP by quarter. Early estimates are revised as more data arrive.

Three US estimates of one quarter arrive in successive releases. The UK also publishes monthly real GDP.

The Fed has eight scheduled policy meetings a year. New data can change rate expectations between meetings.

BEA releases · ONS monthly GDP · FOMC calendar. Snapshot: 16 Oct 2026.

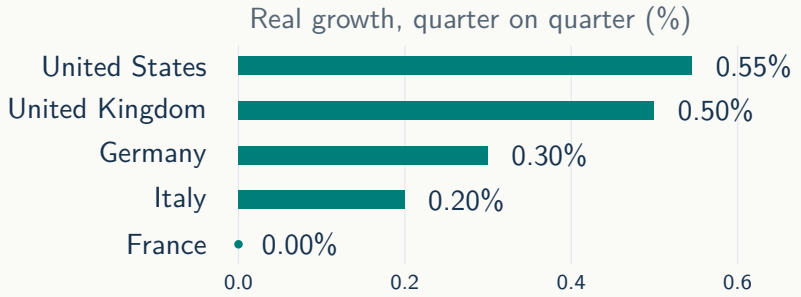
GDP levels: the size of an economy



Annual production at current prices, converted at market exchange rates. USD values also move with currencies; this is not a living-standards comparison.

Trading Economics / World Bank: explore the charts. 2025 data; accessed 16 Oct 2026.

Real GDP growth in Q2 2026



Release date

30 Sep 2026

30 Sep 2026

25 Aug 2026

5 Oct 2026

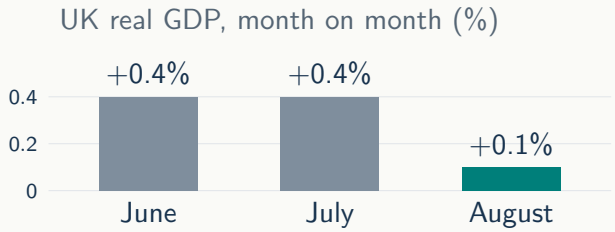
28 Aug 2026

$$g_{\mathrm{US,q/q}} = 100\left[(1.022)^{1/4} - 1\right] \approx 0.55\%.$$

US headline: 2.2% annualized, converted above. All bars are real, seasonally adjusted, nonannualized growth; European data also calendar adjusted.

The latest monthly GDP update

UK release: 15 Oct 2026. Reference month: August 2026.



June–August average GDP: +0.7% versus March–May.

ONS: explore charts and data. Revised estimates.

Italy, 16 Oct: September CPI: +4.2% y/y. Latest GDP (5 Oct): Q2 +0.2% q/q.

GDP news and market expectations

Italy, 30 July 2026: Q2 GDP $+0.2\%$, versus $+0.1\%$ consensus.

$$\underbrace{0.2\% - 0.1\%}_{\text{actual minus consensus}} = +0.1 \text{ percentage points.}$$

Stronger demand may lift profits. It may also raise expected interest rates, lowering bond prices.

Exercise: why might shares fall after good GDP news?

Italy GDP: chart and actual / consensus calendar. A surprise alone does not fix market direction.

LECTURE 1

Accounting and equilibrium

Which equalities always hold, and which need a model?

Expenditure in a closed economy

$$\boxed{Y = C + I + G.}$$

C: private consumption. I: gross domestic investment.

G: government purchases. No international trade yet.

Unsold bread still has an accounting destination

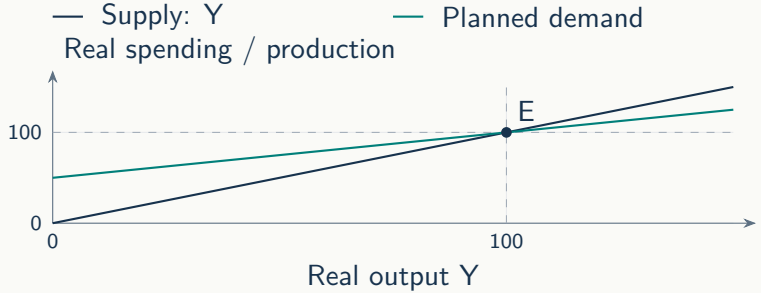
$$Y = 60, \qquad C = 50, \qquad I_{\text{inventory}} = 10.$$

Production is 60. Customers spend 50.

The bakery holds 10 of newly produced unsold bread.

Equilibrium adds planned spending

$$Y = C^{\text{plan}} + I^{\text{plan}} + G.$$



At E, planned spending equals output. Prices are fixed in this model.

Potential output and the long run

$$\boxed{Y^* = AK^{\alpha}(L^*)^{1-\alpha}, \qquad 0 < \alpha < 1.}$$

A: productivity. K: capital. L^*: labour at normal employment.

α: output elasticity of capital in this Cobb–Douglas model.

Potential is sustainable output. The long-run benchmark has $Y = Y^*$.

The output gap connects activity to rates

$$\text{Output gap} = 100\,\frac{Y - Y^*}{Y^*}.$$

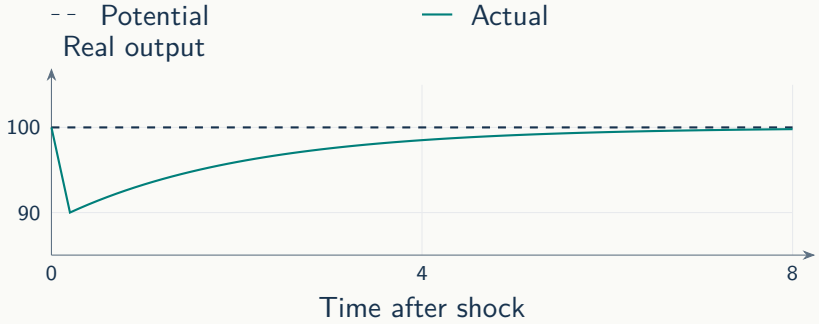
$$Y = 95, \quad Y^* = 100 \quad \Rightarrow \quad \text{gap} = -5\%.$$

Below potential: spare capacity can reduce inflation pressure.

Above potential: stretched resources can increase inflation pressure.

This helps interpret central-bank decisions. The next plot shows a gap closing.

A recession can be an equilibrium with slack



Hypothetical demand shock. The closing gap is a modelled adjustment.

Saving and investment in a closed economy

$$S \equiv Y - C - G.$$

$$Y = C + I + G \quad \Rightarrow \quad \boxed{S = I.}$$

Saving is income not consumed by households or government.

Private and public saving

$$S_p = Y - T - C, \qquad S_g = T - G.$$
$$S = S_p + S_g.$$

T: taxes less current transfers.

A public deficit may be matched by private saving or investment changes.

LECTURE 1

The rest of the world

How do trade, investment income and transfers connect?

Domestic expenditure and trade

$$\boxed{Y = C + I + G + X - M.}$$

X: exports. M: imports. $NX \equiv X - M$.

Imports are removed because spending categories include foreign output.

An imported machine

$$\Delta I = 10, \qquad \Delta M = 10.$$
$$\Delta Y = \Delta I - \Delta M = 0.$$

No extra domestic production at the purchase date in this stripped-down example.

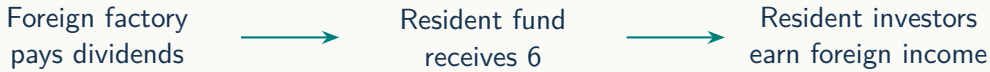
From domestic production to residents' income

$$GNI = Y + NPI.$$

NPI: net primary income received from abroad.

Think interest, dividends, reinvested earnings and cross-border labour income.

Primary income: an ownership example



Income received 6 − income paid abroad $2 = NPI = 4$.

Invented resident/nonresident transactions, in one currency.

Current transfers add disposable income

$$GNDI = Y + NPI + NSI.$$

NSI: net secondary income from abroad.

A resident abroad sending a family remittance is a concrete example.

The current account is our net current flow from abroad

It adds what residents receive from the rest of the world, less what they pay, for trade, income and current transfers.

$$\boxed{CA = (X - M) + NPI + NSI.}$$

A deficit means current payments exceed receipts. The gap needs external financing.

This is why the current account connects trade to saving, investment and funding needs.

Saving finances investment at home or abroad

$$S \equiv GNDI - C - G.$$

$$GNDI = C + I + G + CA.$$

$$\boxed{S - I = CA.}$$

A deficit can finance productive investment

Economy	Saving	Investment	CA
New investment opportunity	20	30	-10
High consumption, same saving gap	10	20	-10

The same current-account deficit can reflect different economic stories.

LECTURE 1

US borrowing and dollar privilege

Why does the world keep financing the United States?

The financial counterpart

$$CA + KA = FA \quad \text{when measured without errors.}$$

KA: capital-account balance.

FA: net acquisition of foreign assets minus net incurrence of liabilities.

Foreign investors help finance US spending



$$CA_{\mathrm{US}} = -\$246.0 \text{ billion in Q2 2026.}$$

BEA, release 24 September 2026. The diagram illustrates one financing channel.

Net international investment position

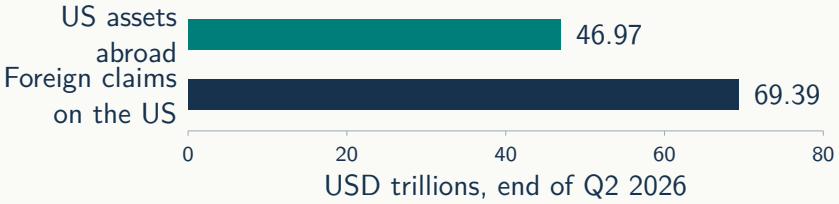
$$\boxed{NIIP = \text{assets held abroad} - \text{foreign claims on us.}}$$

The current account is a flow over a period. NIIP is a balance sheet at a date.

A negative NIIP means foreign claims exceed residents' foreign assets.

Claims include equity and direct investment as well as debt.

The US external balance sheet is large on both sides



$$NIIP = 46.97 - 69.39 = -\$22.42 \text{ trillion.}$$

BEA, end of Q2 2026. These positions include corporate equity and direct investment.

What is the US “exorbitant privilege”?

Investors want dollar assets for reserves, payments and collateral.

They may accept a lower yield for those services, reducing US funding costs.

The US can borrow in dollars while owning riskier businesses abroad.

Dollar funding advantages and returns on the foreign portfolio both matter.

Negative NIIP can coexist with positive investment income

Hypothetical external portfolio	Position	Income yield	Annual income
Foreign assets owned	100	6%	+6
Liabilities to foreigners	150	2%	-3

$$NIIP = 100 - 150 = -50, \qquad \text{net investment income} = 6 - 3 = +3.$$

Illustrative income yields, excluding capital gains. These are not current US observations.

US asset demand connects yields and the dollar

Conditional change	Other inputs fixed
Fewer Treasury buyers	Lower prices, higher yields
Less demand for dollar assets	Pressure on the dollar
A weaker dollar	Higher dollar value of unhedged foreign assets

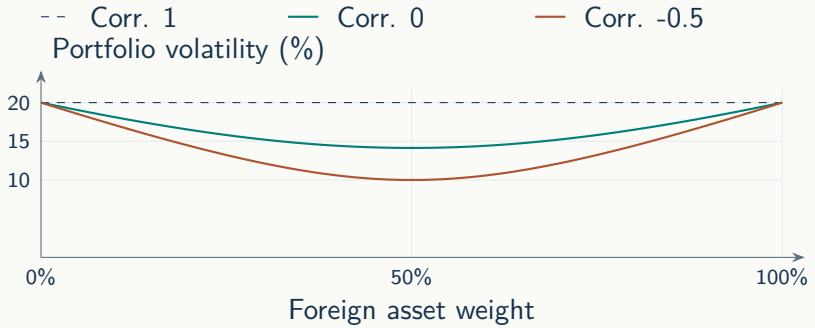
These are conditional mechanisms. NIIP alone gives no market forecast.

Risk sharing is a gain from international finance

Two equally likely states	Home harvest	Foreign harvest
State A	120	80
State B	80	120

Half-ownership of each harvest pays 100 in either state.

Diversification depends on correlation



Equal standalone volatility of 20%. Hypothetical assets and correlations.

More about portfolio diversification in the future.

What a macro release can and cannot tell a trader

GDP describes production relative to expectations and capacity.

The current account describes net external funding needs.

The US example shows why asset demand, income yields and portfolio risk matter.

Counterfactual exercise: extra public consumption

Government consumption rises by 2% of GDP.

Taxes, disposable income and investment initially stay fixed.

Predict the change in national saving and the current account.

The accounting answer and the economic question

$$\Delta S = -2\%, \qquad \Delta CA = -2\% \text{ of GDP}.$$

If output or private saving responds, those numbers change.

Next lecture supplies a model of demand and rates.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.

Money, inflation and activity

How does a policy rate connect spending today to payments tomorrow?

Short run and long run

Short run	Long-run benchmark
Some prices and wages adjust slowly.	Prices and wages have adjusted.
Demand can move output away from potential: $Y \neq Y^*$.	The output gap closes: $Y = Y^*$.
Policy can change real borrowing costs and activity.	Real fundamentals shape the real interest rate.

Purely nominal shocks mainly affect prices in the long-run benchmark. The horizon has no fixed calendar cutoff. Real shocks can move potential output itself.

Money, production and spending



A loan can finance a new oven. Interest rates affect the investment decision.

Today's route

Rates, purchasing power and the long-run real benchmark.

Money demand and classic IS–LM with a fixed money supply.

Inflation feedback, then modern rate targeting at the end.

LECTURE 2

Interest and inflation

What is the price of spending earlier?

A zero-coupon bond exchanges money across time

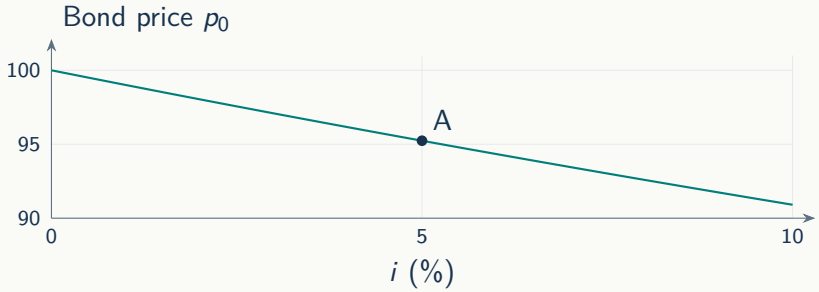
A zero-coupon bond makes one payment at maturity.

$$p_0 = \frac{F}{(1+i)^T}, \qquad F = 100.$$

p_0: price today. i: annual nominal yield. T: years to maturity.

$$T = 1, \quad i = 5\% \quad \Rightarrow \quad p_0 = \frac{100}{1.05} = 95.24.$$

A higher yield means a lower bond price



$$\frac{\partial p_0}{\partial i} = -\frac{TF}{(1+i)^{T+1}} < 0.$$

With face value 100, a one-year price falls from 95.24 at 5% to 90.91 at 10%.

Three rate symbols are enough

Symbol	Meaning
i	Nominal interest rate
$\bar{i}$	Central bank's chosen policy-rate target
r	Real interest rate, measured in purchasing power

Each rate has a stated horizon. In the classic IS–LM model, i is a market outcome.

A real return measures purchasing power

Nominal return counts money. Real return counts what it buys.

$$1 + r_{t+1}^{\text{realised}} = \frac{1 + i_t}{1 + \pi_{t+1}}.$$

Exact purchasing-power conversion: return and realised inflation cover the same period.

The Fisher approximation comes from logarithms

$$\ln(1 + r) = \ln(1 + i) - \ln(1 + \pi).$$

For small rates, $\ln(1 + x) \approx x$.

$$\boxed{r \approx i - \pi.}$$

Exact in log gross rates; approximate in simple percentage rates. This line uses realised inflation.

For decisions today, expected inflation matters

$$\boxed{r_t \approx i_t - \pi_t^e, \qquad \pi_t^e \equiv \mathbb{E}_t \pi_{t+1}.}$$

π_t^e: expected inflation over the investment horizon.

$$i_t = 5\%, \quad \pi_t^e = 3\% \quad \Rightarrow \quad r_t \approx 2\%.$$

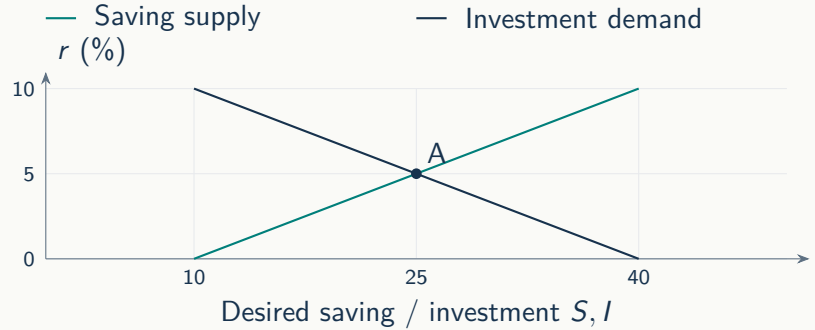
From here, r denotes the approximate expected real rate relevant for decisions.

A nominal cut need not reduce the real rate

	Before	After
Nominal rate i	5%	4%
Expected inflation	3%	1%
Approximate real rate r	2%	3%

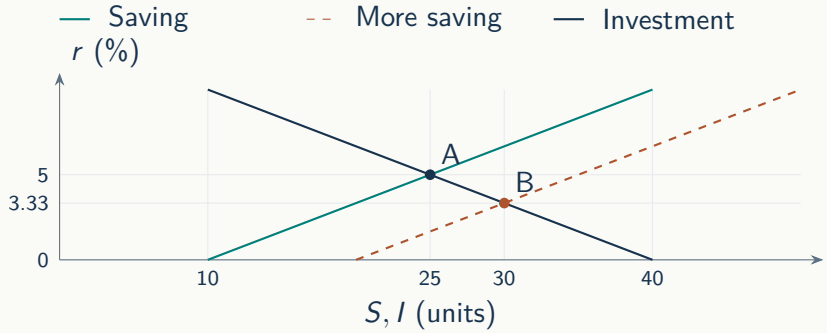
Expected inflation fell faster than the nominal rate.

Saving, investment and the real rate



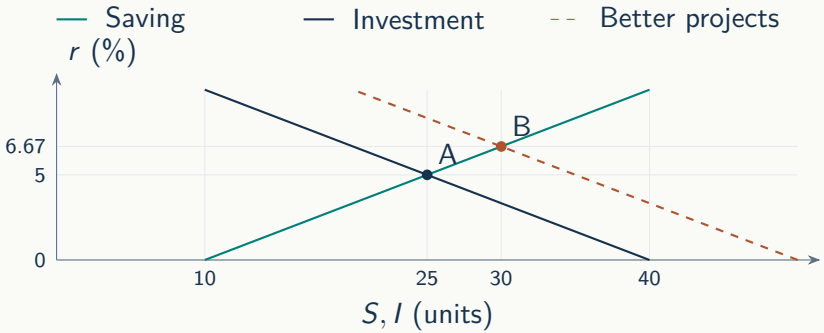
At A: planned saving and investment both equal 25; r = 5%. Hypothetical units.

A larger harvest can lower the real rate



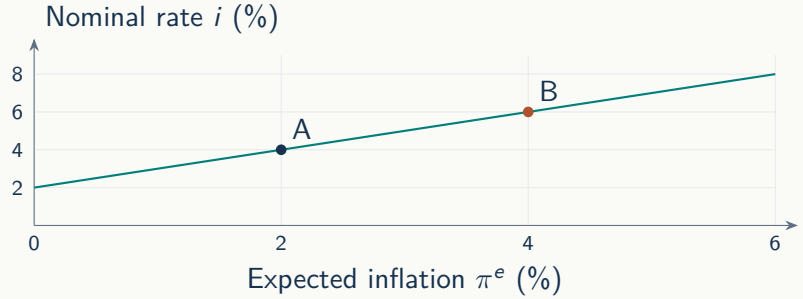
Extra income raises saving supply by 10. The real rate falls from 5% to 3.33%.

Better projects can raise the real rate



Firms want 10 more units at each rate. Competition for resources raises r from 5% to 6.67%.

The long-run Fisher effect



$$i \approx r^* + \pi^e, \qquad r^* = 2\% \text{ held fixed.}$$

With r* fixed, expected inflation and the nominal rate both rise by about 2 points.

The causal story depends on the horizon

Long-run benchmark	Short-run experiment
Resources, tastes and technology pin down the real benchmark. $r = r^*, \quad i \approx r^* + \pi^e$.	Sticky prices and expectations can adjust slowly. Policy changes $\bar{i}$; then $r \approx \bar{i} - \pi^e$ changes.

A higher nominal rate can reflect higher expected inflation or tighter real restraint. Specify the experiment.

Inflation measures a change in the price level

P_t: the cost index of a representative basket.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Inflation means the same money buys less of that basket.

Where does the extra money for higher prices come from? The next example needs none.

Prices can rise without any extra money

One-good economy, one period	Before	Supply shortfall
Money spent in the period	100	100
Goods available	10	5
Price per good	10	20

The same spending competes for fewer goods.

Fixed nominal spending and no extra money in this hypothetical example.

New money and faster spending are other possibilities

$$MV = PY, \qquad V \equiv \frac{PY}{M}.$$

Bank lending can create new deposits.

Existing balances can circulate faster: the same money supports more spending.

M is a money stock; PY is nominal output over a period; V is its turnover ratio.

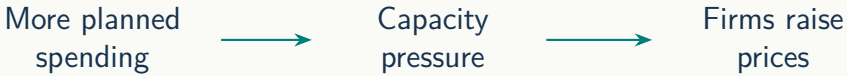
Disinflation can leave prices higher

Date	Basket price	Inflation
0	100	–
1	110	10%
2	112.20	2%

Inflation fell. The price level continued to rise.

Demand pressure can raise prices

Orders grow faster than firms can expand production.



The effect depends on how much productive slack remains.

Supply shocks can raise prices while output weakens

An energy shortage makes production more expensive.

Firms may raise prices and produce less at the same time.

A central bank can influence demand. It cannot create the missing energy.

Expectations can make inflation persistent

Expected cost increases enter wage negotiations and pricing plans.

Contracts can carry an initial shock into later periods.

Wage catch-up alone does not establish an explosive wage-price spiral.

Why high or volatile inflation is costly

Unindexed income and savings lose purchasing power.

Unexpected inflation redistributes value in nominal contracts.

Unstable prices make planning and relative-price signals harder.

Why choose a small positive target?

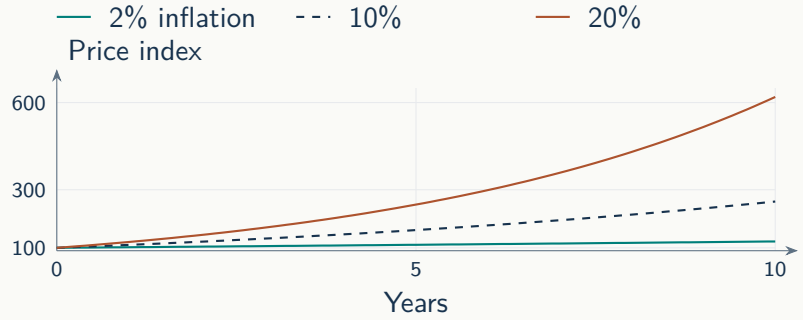
A buffer against persistent deflation.

More room to lower real rates when nominal rates face a lower bound.

Easier real-wage adjustment when nominal wages are sticky.

The ECB targets 2% inflation over the medium term. This is a policy choice.

Constant inflation compounds the price level



$$P_t = P_0(1+\pi)^t.$$

LECTURE 2

Money and short rates

What can a central bank actually change?

A bank loan creates an asset and a deposit

Bank assets	Bank liabilities
Loan to borrower: +100	Borrower's deposit: +100

The borrower receives spendable money and an equal debt.

Lending still faces credit, capital, funding and liquidity constraints.

The central bank anchors short-term market rates

Banks use central-bank reserves to settle payments.

Administered rates and operations implement the overnight target $\bar{i}$.

Long yields also depend on expected future policy and risk premia. Exceptional regimes target them directly.

Exception: a central bank can target a long yield

In 2016, the BoJ introduced yield curve control.

It bought Japanese government bonds to keep the ten-year yield around 0%.

$$p_0 = \frac{100}{(1+i)^T}: \quad \text{support the bond price to restrain its yield.}$$

A historical policy example. The ZCB formula illustrates the price/yield mechanism.

Money is useful, but costly to hold

Money settles purchases and provides liquidity.

Holding it can give up interest on an alternative asset.

Real money balances measure purchasing power

$$\text{Real balances} = \frac{M}{P}.$$

A representative basket	Before	Prices double
Nominal money M	100	100
Price P per basket	10	20
Baskets the balance can buy	10	5

M/P is purchasing power held for transactions, measured as a stock.

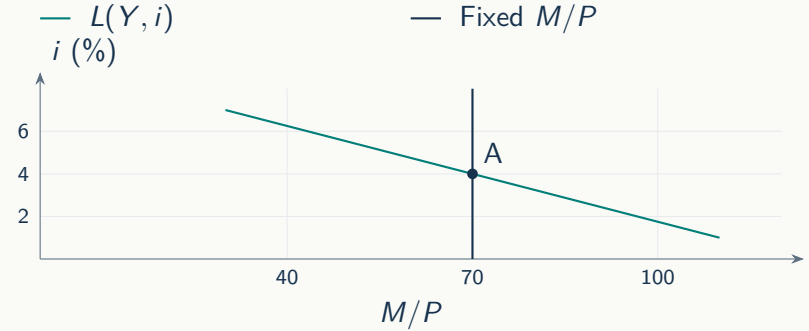
The money-market condition

$$\frac{M}{P} = L(Y, i), \qquad L_Y > 0, \quad L_i < 0.$$

Symbol	Meaning
M	Nominal money stock chosen in the classic model
P	Price level, fixed in this experiment
L(Y,i)	Desired real money holdings

More real transactions raise money demand. Higher bond interest makes cash more costly to hold.

The money-demand diagram



Classic money market: fixed prices P and real output Y. Hypothetical units.

LECTURE 2

IS–LM and shocks

What changes after spending or policy changes?

What is an economic model?

Like a model in physics: isolate a mechanism and state what is fixed.

For now, we care about slopes, derivatives, intersections and shifts.

Specific numbers and functional forms help draw examples; they are assumptions.

The aim here is economic intuition. Quantitative forecasts later require measurement and estimation.

The variables and assumptions of this model

Variable	Meaning
Y	Real output and the income generated
i	Nominal market short rate
P	Price level, fixed here

Expected inflation and credit spreads are fixed. Compare short-run equilibria, possibly below potential.

A firm needs 10 million to start a project

Today: 8 million borrowed + 2 million of the owner's funds.

One year later: the project pays 15 million.

$$\text{Debt repayment} = 8(1 + i).$$

The owner compares the project with an alternative

$$\text{Equity payoff} = 15 - 8(1 + i).$$

$$\text{Alternative payoff} = 2(1 + i).$$

Cash left after repaying debt must also compensate the owner for the alternative investment.

The project's net present value

$$NPV = \frac{15 - 8(1+i)}{1+i} - 2$$

$$\boxed{NPV = \frac{15}{1+i} - 10.}$$

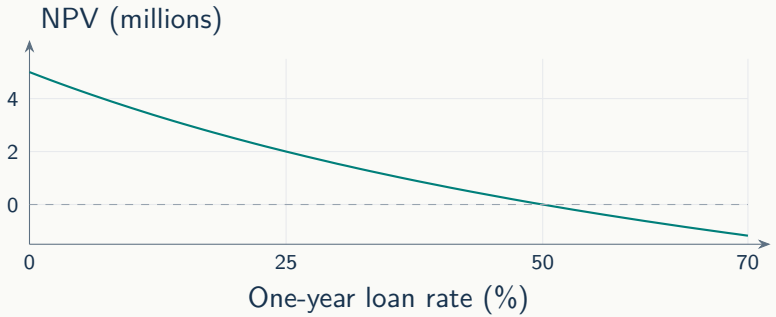
NPV is value today after subtracting today's investment.
Both expressions use the same certainty and discount-rate assumptions.

Working through the financing cost

Loan rate	Debt repayment	Equity payoff	Project NPV
5%	8.40	6.60	4.29
25%	10.00	5.00	2.00
50%	12.00	3.00	0.00
60%	12.80	2.20	−0.63

All cash amounts are millions. At 60%, the alternative equity payoff is 3.20.

Higher rates eliminate marginal projects



Each project has a threshold. A collection of projects creates an investment response.

IS means Investment–Saving

IS lists the output/rate pairs where
planned spending equals production.

In a closed-economy equilibrium,
planned investment equals planned national saving.

Why IS slopes downward

A higher rate makes fewer investments worthwhile.

Lower investment means less planned expenditure.

Equilibrium output is lower.

Output is on the horizontal axis. The rate is on the vertical axis.

Households respond to disposable income

$$C = C_0 + c(Y - T), \qquad 0 < c < 1.$$

Symbol	Meaning
C	Planned consumption
C_0	Consumption independent of current income
T	Taxes less transfers
c	Extra consumption per extra unit of disposable income

Investment responds to the rate

$$I = I_0 - bi, \qquad b > 0.$$

Symbol	Meaning
I	Planned investment
I_0	Spending independent of the current rate
b	Response to the nominal rate, with other inputs fixed

Fixed expected inflation and credit spreads make i a proxy for the relevant real financing cost.

Combining spending gives the IS equation

G: government consumption, fixed in this experiment.

$$Y = C + I + G.$$

$$\boxed{Y = \frac{C_0 - cT + I_0 + G - bi}{1 - c}.}$$

The equation records all goods-market equilibria for the chosen assumptions.

LM means Liquidity preference–Money supply

LM lists output/rate pairs where
desired money balances equal money supply.

Holding money is useful for payments.
Holding it also gives up interest on other assets.

Why classic LM slopes upward

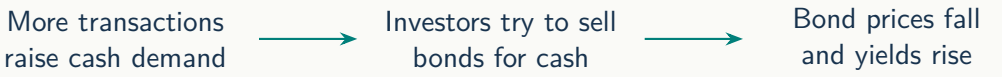
$$\frac{M}{P} = L(Y, i), \qquad \frac{di}{dY} = -\frac{L_Y}{L_i} > 0.$$

More output means more payments and a greater desire to hold money.

With M/P fixed, i rises until the opportunity cost of cash offsets that extra demand.

The central bank chooses M. Private money demand and the bond market determine the clearing i.

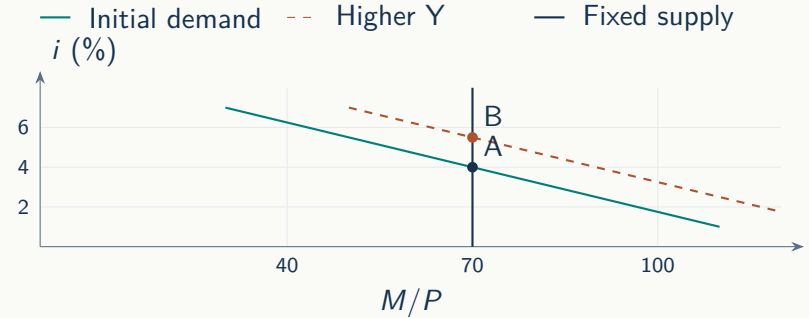
Agents adjust portfolios, and bond prices move



The total money stock is unchanged. Higher i reduces desired cash holdings.

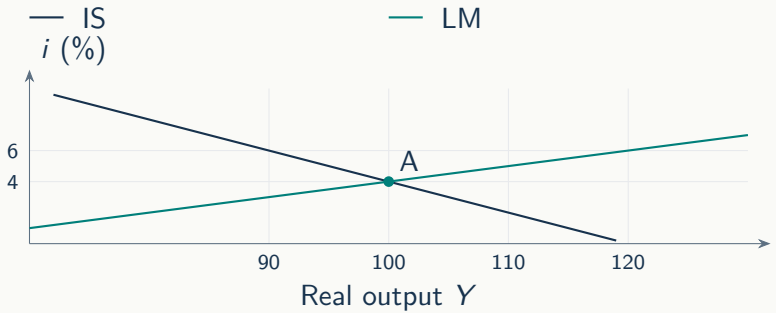
It is a portfolio opportunity-cost mechanism. Spending need not disappear to clear the money market.

Higher output shifts money demand to the right



At the same fixed M/P, higher real activity raises the clearing market rate.

The intersection balances both markets



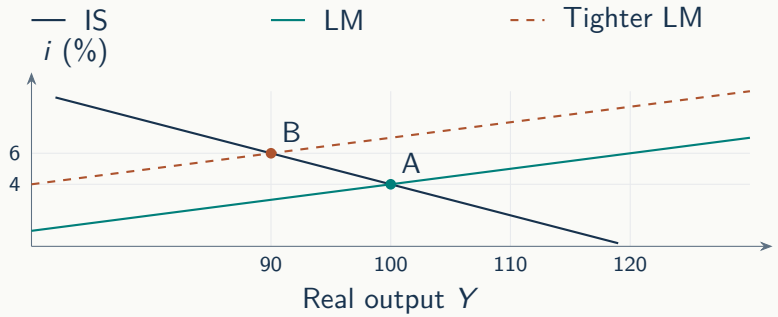
At A, planned expenditure equals output and money demand equals supply.

A movement along a curve versus a shift

Movement along IS	**A shift of IS**
A change in the rate, holding other spending inputs fixed.	A change in spending at each given rate.

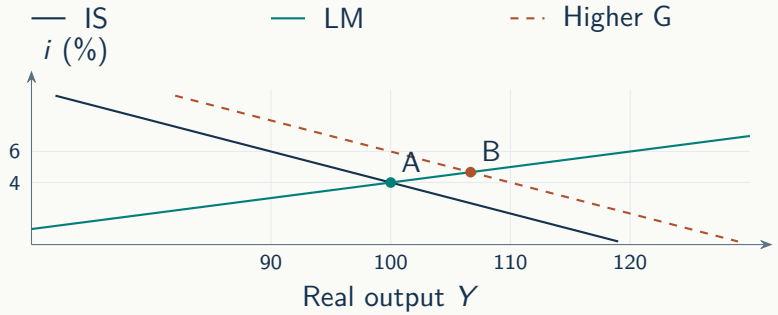
First identify what changed. Then identify which relationship changes.

Monetary tightening shifts classic LM



Lower real money supply: A becomes B. The rate rises and output falls.

Fiscal expansion with an unchanged money supply



The same IS shift raises output and the rate. Private investment partly falls.

A demand shock versus a supply shock

Lower confidence shifts planned spending inward.

Lower productive capacity moves potential output.

Policy faces different inflation consequences in the two cases.

LECTURE 2

Inflation and policy feedback

Why can a central bank raise nominal rates by more than inflation?

The neutral real rate is a benchmark

r^*: the real rate consistent with output at potential and stable inflation.

The actual real rate may be above or below it.

Neutral is an estimate. It depends on the economy's resources, preferences and opportunities.

At the target, a neutral nominal benchmark

$$\bar{i} \approx r^* + \pi^*.$$

π^*: the inflation target.

$$r^* = 1\%, \quad \pi^* = 2\% \quad \Rightarrow \quad \bar{i} \approx 3\%.$$

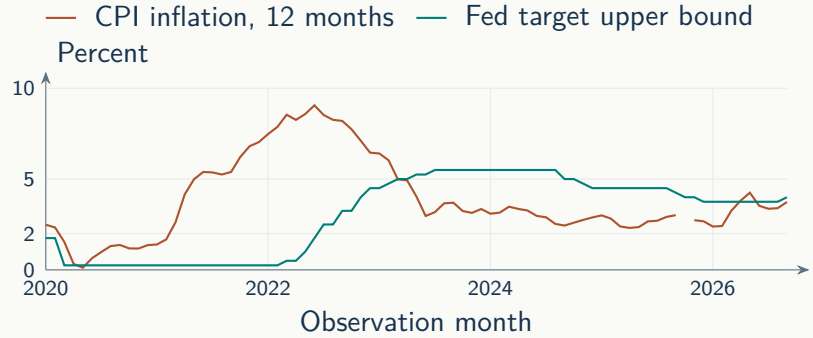
If inflation rises, the nominal rate must catch up

Suppose actual and expected inflation rise from 2% to 3%.

$$\text{To keep a 1\% real rate:} \quad \bar{i} = 4\%.$$

To add restraint, an illustrative rule raises it further to 4.5%.

US inflation and policy after Covid



Jan 2020–Sep 2026. BLS/Federal Reserve via FRED. Policy: month-end. Missing Oct 2025 CPI is a gap.

The output gap measures pressure on resources

$$x_t = 100 \frac{Y_t - Y_t^*}{Y_t^*}.$$

Symbol	Meaning
Y_t	Actual real output
Y_t^*	Potential real output
x_t	Output gap in percent

Positive gap: output above potential. Negative gap: slack.

Weak activity changes the policy trade-off

Keep inflation at 3%, but put output 2% below potential.

$$\text{Illustrative response:} \quad 4.5 - 0.5 \times 2 = 3.5\%.$$

Inflation calls for restraint. Slack calls for support.

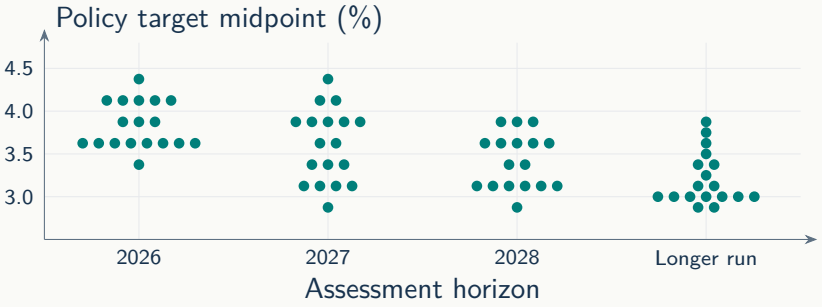
A central banker might follow a rule like this

$$\bar{i}_t = r^* + \pi_t + 0.5(\pi_t - \pi^*) + 0.5x_t.$$

One possible policy reaction: respond to inflation and the output gap.

Illustrative Taylor-style rule. Rates and gaps use percentage points; this is not a commitment.

Fed policy assessments: June 2026



Official medians, rounded: 3.8%, 3.6%, 3.4%, 3.1%. Federal Reserve SEP, 17 June 2026, Figure 2.

What each dot means

One official's assessment of appropriate policy at that year-end.

The assessment depends on that official's economic outlook.

Dots are anonymous. We cannot join a person's path across columns.

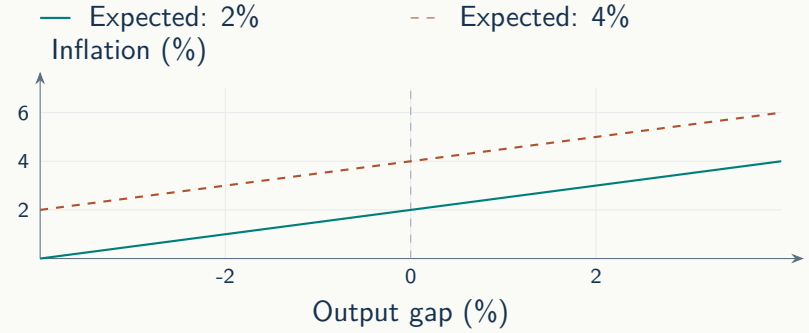
The dispersion is across officials. It is not a probability distribution of future policy outcomes.

A simple Phillips relationship

$$\pi_t = \pi_t^e + \kappa x_t + u_t.$$

Symbol	Meaning
π_t^e	Expected inflation for this period
x_t	Output gap, as defined earlier
$\kappa > 0$	Inflation response to demand pressure
u_t	Cost or supply disturbance

Expectations shift the Phillips curve



Hypothetical $\kappa = 0.5$ and no supply disturbance.

Inflation deviations are easier to analyse

$$d_t = \pi_t - \pi^*.$$

Symbol	Meaning
d_t	Inflation deviation from target, in percentage points
π_t	Inflation rate
π^*	Inflation target

Inflation of 3% with a 2% target means $d_t = 1$.

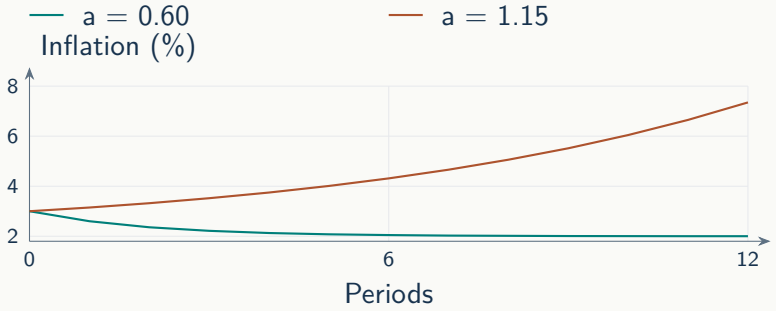
A simple model of inflation persistence

$$d_t = a\, d_{t-1} + u_t.$$

Symbol	Meaning
a	Net carry-over from the previous inflation deviation
u_t	A new inflation disturbance this period

After an initial shock, set later disturbances to zero and compare values of a.

The same shock can decay or amplify



Target 2%, initial deviation 1 point. Stable when $|a| < 1$. Original hypothetical paths.

Exercise: inflation rises while output falls

An energy shortage raises costs and reduces productive capacity.

Compare holding the nominal rate fixed with raising it.

Which assumptions determine the change in the real rate?

LECTURE 2

Modern rate targeting

What changes when the central bank chooses the rate?

Choose the rate, and let the monetary quantity adjust

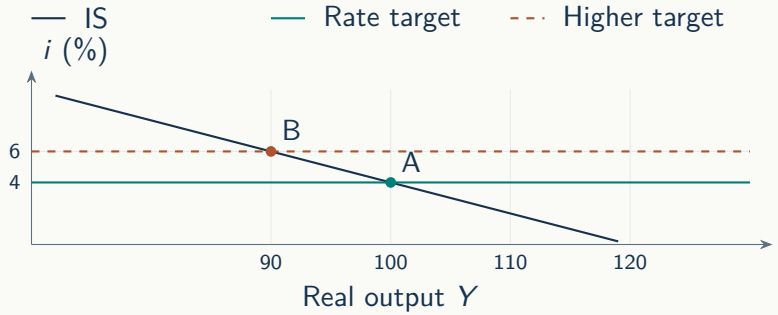
$$i = \bar{i}, \qquad \frac{M}{P} = L(Y, \bar{i}).$$

The central bank chooses the target. It accommodates the quantity needed to implement it.

For this experiment, the target is fixed as Y changes.

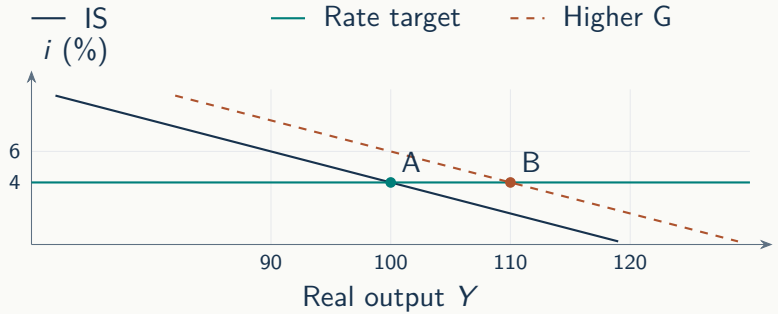
Across meetings, the chosen target can respond to inflation and activity.

With a rate target, policy sets a horizontal line



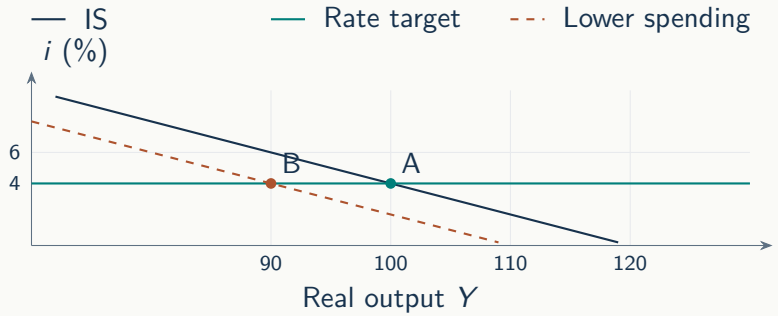
Chosen $i = \bar{i}$ rises from 4% to 6%; output falls from 100 to 90.

Fiscal expansion with an unchanged rate target



Higher government spending shifts IS outward. At a fixed rate, output rises.

A private-demand slowdown shifts IS inward



Lower autonomous consumption or investment reduces output at a fixed target.

LECTURE 2

Optional mathematical extension

How does policy change an inflation recurrence?

Optional extension: what changes the feedback?

We will connect three mechanisms:

Inflation pressure, the demand response to real rates, and a policy response to inflation.

Each relationship comes next on its own slide. The coefficients describe this toy model only.

Inflation responds to activity and past inflation

$$d_t = \alpha d_{t-1} + \kappa x_t + u_t.$$

New coefficient	Meaning
α	Carry-over of the previous inflation deviation
$\kappa > 0$	Inflation response to a one-point output gap

d is the inflation deviation, x the output gap, and u a new disturbance.

Real-rate restraint reduces the output gap

$$x_t = -B(r_t - r^*), \qquad B > 0.$$

B: how strongly demand responds to the real-rate gap.

Higher real rates relative to neutral reduce demand.
B is an aggregate sensitivity, distinct from investment's earlier coefficient b.

Policy adds real restraint when inflation exceeds target

$$r_t - r^* = \phi d_{t-1}, \qquad \phi > 0.$$

ϕ: the real-rate response to a one-point inflation deviation.

The nominal policy rate must also offset expected inflation to deliver this real response.

Substitution gives a stability condition

$$d_t = \underbrace{(\alpha - \kappa B \phi)}_{a} \, d_{t-1} + u_t.$$

$$\boxed{|\alpha - \kappa B \phi| < 1.}$$

If $\alpha = 1.15$ and $\kappa B \phi = 0.55$, then $a = 0.60$.

Policy supplies negative feedback in this specified model.

A two-date endowment economy

$$U = \ln c_0 + \beta \ln c_1.$$

c_0, c_1: consumption now and later. $\beta > 0$: patience.

e_0, e_1: the economy's available resources on the two dates.

U: lifetime utility. The real bond transfers purchasing power between dates.

The individual intertemporal budget

$$c_0 + \frac{c_1}{1+r} = e_0 + \frac{e_1}{1+r}.$$

Saving one unit now buys a claim to future resources.

The individual chooses consumption given the rate.

Market clearing pins down a real rate

$$\frac{1}{c_0} = \beta(1+r)\frac{1}{c_1}, \qquad c_0 = e_0, \quad c_1 = e_1.$$

$$\boxed{1 + r = \frac{e_1}{\beta e_0}.}$$

More resources now lower the rate, other inputs fixed.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.

ECONOMICS AND FINANCE / 03

Bonds and yield curves

Pricing cash flows, measuring risk



Today's route

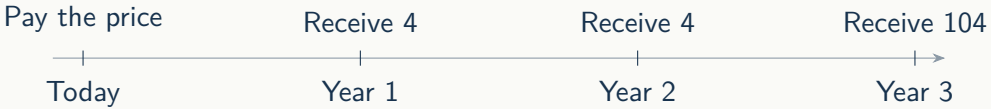
Section	Question
Cash flows and prices	What exactly are we buying?
Bills and auctions	How does a new security get priced?
Curves and forwards	What can we lock in today?
Expectations and term premia	Which part of a yield is a forecast?
Duration and convexity	How much do prices move?
Hedging and market cases	What position expresses the view?

LECTURE 3

Cash flows and prices

What is promised, what is priced, and what can we earn?

A bond is a dated payment contract



$$F = 100, \qquad c = 4\%, \qquad C = cF = 4.$$

Prices and cash-flow symbols

Symbol	Meaning
F	Face value, normally 100 in our examples
$p_t(T)$	Bond price at time t, with T years remaining
$C = cF$	Coupon payment from coupon rate c
$d_t(T)$	Price today of one certain unit paid in T years

Rate symbols and horizons

Symbol	Meaning
$i_t(T)$	Nominal spot rate for a T-year zero
$f_t(T_1, T_2)$	Rate fixed today for a future lending interval
i^{YTM}	One yield matching a coupon bond's price
$\bar{i}_t$	Central bank's chosen short-rate target

r still denotes a real rate. Pricing examples use nominal rates in decimals.

A zero-coupon bond

$$p_t(T) = \frac{F}{[1 + i_t(T)]^T}.$$

$$F = 100, \quad T = 2, \quad i_t(2) = 5\% \quad \Rightarrow \quad p_t(2) = 90.70.$$

A discount factor prices one future unit

$$d_t(T) = \frac{p_t^{\text{zero}}(T)}{F} = \frac{1}{[1 + i_t(T)]^T}.$$

$$p_t = \sum_{k=1}^{n} d_t(k)\, CF_k.$$

CF_k: the cash payment at date k. Match dates, currency and credit risk.

Replication gives the pricing intuition

Buy the same cash flows using zero-coupon bonds.

A matching payoff schedule has a matching price.

Same currency, payment dates and credit quality. Borrowing, shorting and trading must be feasible.

A coupon bond is a basket of zeros

Date	Cash flow	Replication
1	4	4 units of a date-1 zero
2	4	4 units of a date-2 zero
3	104	104 units of a date-3 zero

$$p_t = 4d_t(1) + 4d_t(2) + 104d_t(3).$$

Worked price with a flat curve

$$p_t = \frac{4}{1.05} + \frac{4}{1.05^2} + \frac{104}{1.05^3} = 97.28.$$

A 4% coupon trades below face value when the required yield is 5%.

Annual coupons. Flat 5% spot curve. Valuation on a coupon date.

Market price determines yield to maturity

$$p_t = \sum_{k=1}^{n} \frac{CF_k}{(1 + i^{\mathrm{YTM}})^k}.$$

Given the price and promised payments, solve for the single rate.

YTM summarizes the bond price. Each dated spot rate can still be different.

Coupon, yield and return

Object	What does it measure?
Coupon rate c	Contractual annual payment per unit face
Yield i^{YTM}	Discount rate matching today's price
Holding return H	Cash received plus price change, over your horizon

$$H_{t,t+1} = \frac{C_{t+1} + p_{t+1} - p_t}{p_t}.$$

A 5% yield can produce a one-year loss

Ten-year zero, face 100	Price
Buy at 5% today	61.39
Sell one year later, remaining nine-year yield 6%	59.19

$$H = \frac{59.1898 - 61.3913}{61.3913} = -3.59\%.$$

What holding to maturity locks in

A default-free zero locks its nominal payment and annualized return.

Coupon-bond wealth also depends on coupon reinvestment.

Selling early introduces price risk. Inflation and foreign-currency returns add other risks.

Risk-free depends on the investor's horizon

Need	Relevant risk
100 in ten years	A matched ten-year zero fixes that nominal payment
Spend after one year	A long zero has uncertain resale value
Roll bills for ten years	Future bill reinvestment rates are unknown

LECTURE 3

Bills and auctions

How does a new Treasury security get priced?

Treasury bills are short zero-coupon claims

No periodic coupon. The investor receives face value at maturity.

Regular original terms	Minimum purchase
4, 6, 8, 13, 17, 26 and 52 weeks	100 dollars

TreasuryDirect rules checked 16 October 2026. Cash-management bills can have other terms.

How the auction clears

Treasury announces the security, offering size and dates.

Eligible noncompetitive bids receive their requested amounts.

Competitive bids enter from low rates to high rates until the offering is allocated.

Two ways to submit a bid

Noncompetitive	Competitive
Specify the face amount.	Specify face amount and required rate.
Accept the auction's clearing rate.	A bid can be accepted, partly filled or rejected.

Successful bidders pay the uniform auction price.

An invented bill auction

Bidder	Requested face (millions)	Discount-rate bid
A	30	3.90%
B	30	4.00%
C	30	4.10%
D	20	4.20%

Offering: 100 million. Noncompetitive bids: 20 million. Competitive allocation remaining: 80 million.

The cutoff bidder receives a partial fill

Bidder	Accepted face (millions)	Outcome
A	30	Full
B	30	Full
C	20 of 30	Two-thirds
D	0	Rejected

Stop-out discount rate $= 4.10\%$.

One price for every successful bidder

$$p = F\left(1 - i_d \frac{N}{360}\right).$$

$$F = 100, \quad i_d = 4.10\%, \quad N = 28 \quad \Rightarrow \quad p = 99.681111.$$

i_d: bank-discount quotation. N: calendar days from issue to maturity.

Actual four-week auction: 15 October 2026

Reported item	Official result
Public offering	110 billion dollars
High discount rate	4.000%
Price per 100 face	99.688889
Reported bid-to-cover	2.56

Issue: 20 October. Maturity: 17 November. CUSIP 912797VX2. Source: TreasuryDirect.

The same bill has different rate quotations

$$H = \frac{F - p}{p} = 0.3121\% \quad \text{over 28 days.}$$

$$i_d = \frac{F - p}{F}\frac{360}{28} = 4.000\%.$$

$$i_{\text{inv}} = \frac{F - p}{p}\frac{365}{28} = 4.068\%.$$

Same cash flow. Different denominators and annualization rules.

Coupon bonds also have clean and dirty prices

$$p_{\text{dirty}} = p_{\text{clean}} + \text{accrued interest.}$$

$$99.20 + 0.80 = 100.00 \quad \text{per 100 face.}$$

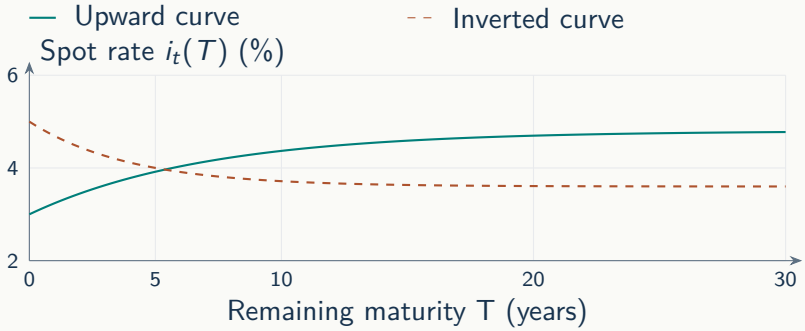
The dirty price is the cash exchanged at settlement.

LECTURE 3

Curves and forwards

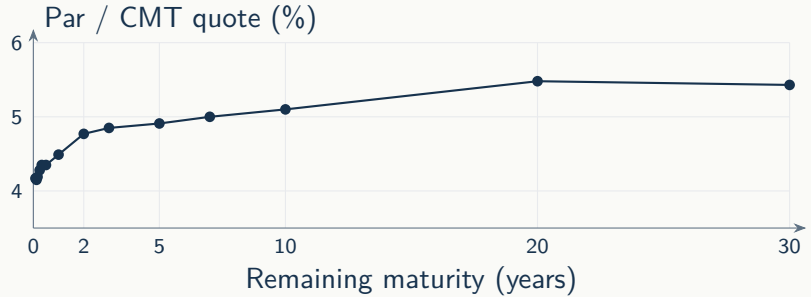
What do prices at different maturities let us lock in?

A yield curve is one date, many maturities



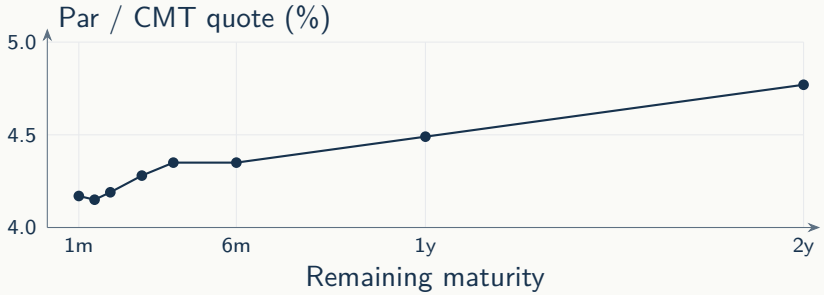
Hypothetical shapes. The horizontal axis is maturity, not the future calendar path of policy.

US Treasury par curve: all published tenors



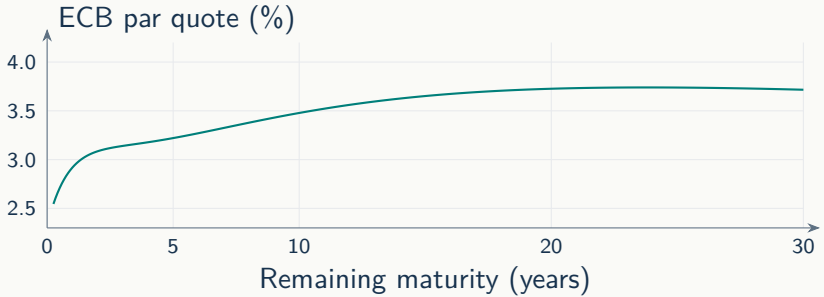
15 October 2026. All 14 published tenors, including the bill end. Treasury quotation conventions.

The short end deserves a closer look



Same 15 October observations. A zoom changes visibility, not the data.

Euro-area AAA: 358 fitted maturities



14 October 2026. Published monthly grid, 3 months to 30 years; fitted points are not independent quotes.

The zero-coupon (spot) yield curve

One payment at maturity T, with no intermediate coupons.

$$i_t(T) = \left[\frac{F}{p_t^{\text{zero}}(T)}\right]^{1/T} - 1.$$

The curve plots this annualized rate against maturity T.

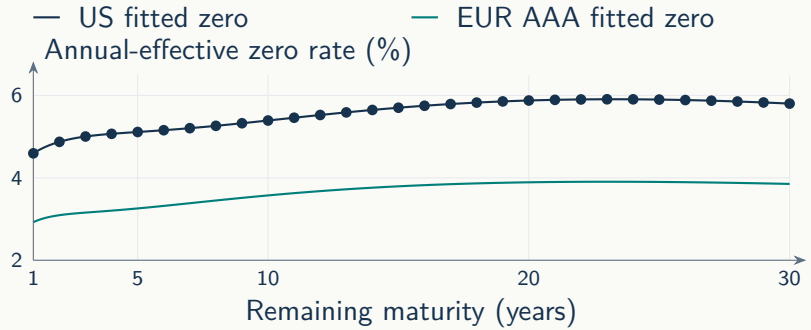
Each point discounts a single payment: $p_t^{\text{zero}}(T) = F[1 + i_t(T)]^{-T}$.

STRIPS and estimated spot rates

Object	Meaning
Treasury STRIP	A traded principal or coupon payment separated from a Treasury security.
Fitted ZCB curve	Discount factors inferred by fitting observed bond prices.

The US panel fits off-the-run coupon bonds. Its points are model-implied ZCB yields, rather than STRIPS quotes.

Fitted zero-coupon spot yields



9 October 2026. Continuous source yields converted by $i = \exp(z) - 1$. Different currencies and samples.

Spot, par and YTM are different objects

Rate	Pricing question
Spot $i_t(T)$	What discounts one payment at T?
Par coupon rate	Which coupon makes a new bond worth F?
Bond-specific i^{YTM}	Which single rate fits this coupon schedule's price?

More plotted maturities make the fitted shape visible. They do not add independent market quotes.

A non-flat curve discounts dates differently

$$p_t = \frac{C}{1 + i_t(1)} + \frac{C + F}{[1 + i_t(2)]^2}.$$

Use each payment's own spot rate. The coupon rate does not choose the discount rate.

Bootstrapping a two-year discount factor

$$d_t(1) = 0.96, \quad C = 5, \quad F = 100, \quad p_t = 100.$$

$$100 = 5(0.96) + 105 d_t(2) \quad \Rightarrow \quad d_t(2) = 0.906667.$$

$$i_t(2) = d_t(2)^{-1/2} - 1 = 5.021\%.$$

A forward loan is agreed today



The future loan rate is fixed before the money changes hands.

Two ways to lock a two-year payoff

Route A	Route B
Buy a two-year zero.	Invest for one year.
Lock the whole two-year return today.	Lock year 2 using a forward agreed today.

Same initial investment, final date, currency, credit and collateral assumptions.

Exact no-arbitrage across the two routes

$$\boxed{[1 + i_t(2)]^2 = [1 + i_t(1)][1 + f_t(1, 2)].}$$

The forward rate adjusts until the locked final payoffs match.

Worked forward rate

$$i_t(1) = 4\%, \qquad i_t(2) = 5\%.$$

$$f_t(1,2) = \frac{1.05^2}{1.04} - 1 = 6.0096\%.$$

$$100(1.05)^2 = 100(1.04)(1.060096) = 110.25.$$

Replicating the forward with two zeros

Position	Today	Year 1	Year 2
Short 100-face one-year zero	$+96.1538$	-100	0
Buy 106.0096-face two-year zero	-96.1538	0	$+106.0096$
Net	0	-100	$+106.0096$

A future loan is a portfolio of dated claims. F is face value; f is the forward rate.

A mispriced forward creates a locked profit

Suppose an otherwise identical quoted forward lets you lend at 8%.

Borrow synthetically at 6.0096%, then lend the same 100 at 8%.

$$\text{Year-2 net cash flow} = 108 - 106.0096 = 1.9904.$$

A forward is a price; a forecast is a belief

Known today	Still uncertain
Locked $f_t(1,2) = 6.0096\%$	The one-year spot rate observed next year.
Its value follows from today's discount factors.	Its expectation needs a model or survey.

No-arbitrage links traded prices. It does not set the physical probability of future outcomes.

LECTURE 3

Expectations and term premia

How much of a long yield can we interpret as a forecast?

Rolling over leaves the second rate unknown

Year-2 rate after a 4% first year	Final wealth from 100
2%	$100(1.04)(1.02) = 106.08$
8%	$100(1.04)(1.08) = 112.32$
Two-year zero locked at 5%	110.25

An unhedged roll-over is risky. The forward-locked strategy on earlier slides is different.

Exact compounding across maturities

$$[1 + i_t(n)]^n = \prod_{k=0}^{n-1} [1 + f_t(k, k+1)].$$

$$\ln[1 + i_t(n)] = \frac{1}{n} \sum_{k=0}^{n-1} \ln[1 + f_t(k, k+1)].$$

$f_t(0, 1) = i_t(1)$. These are rates locked today, not expected future spot rates.

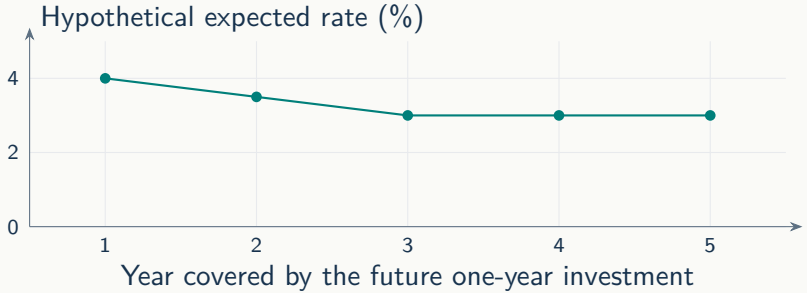
A premium for each forward interval

$$\mu_k \equiv \mathbb{E}_t[i_{t+k}(1)], \qquad f_t(k, k+1) = \mu_k + \varphi_t(k).$$

$\varphi_t(k)$ is the forward-minus-expected-rate wedge for year k.

It can differ by horizon and observation date, and can have either sign.

An explicit expected short-rate path



Hypothetical expectations: 4.0%, 3.5%, 3.0%, 3.0%, 3.0%. A forecast does not lock a return.

Why a term premium can appear

A short-horizon holder of a long bond faces resale-price risk.

Investors may demand compensation for exposure to bad outcomes.

A bond that hedges the investor's bad states can instead command an insurance value.

Premiums can differ across future years

Year k	Expected rate	Forward premium	Locked forward
0	4.0%	0.0 points	4.0%
1	3.5%	0.4 points	3.9%
2	3.0%	0.6 points	3.6%
3	3.0%	0.8 points	3.8%
4	3.0%	1.0 points	4.0%

Hypothetical. In each row, $f_t(k, k+1) = \mu_k + \varphi_t(k)$.

Compound the forward rates, period by period

$$\boxed{[1 + i_t(n)]^n = \prod_{k=0}^{n-1} [1 + \mu_k + \varphi_t(k)]\,.}$$

$$\mu_k = \mathbb{E}_t[i_{t+k}(1)], \qquad \varphi_t(k) = f_t(k, k+1) - \mu_k.$$

The premium can vary in every future period. Gross returns multiply across periods.

Worked five-year compounded yield

$$(1 + i_t(5))^5 = 1.040 \times 1.039 \times 1.036 \times 1.038 \times 1.040.$$

$$i_t(5) = (1.040 \times 1.039 \times 1.036 \times 1.038 \times 1.040)^{1/5} - 1 = 3.8599\%.$$

With all future premiums set to zero, the same expected path gives 3.2992% under that added assumption.

Three objects to keep separate

Object	Whose information?	What is it?
Fed dots	Individual officials	Conditional policy assessments
Forward $f_t(T_1, T_2)$	Market prices today	Rate on a locked future interval
Expectation $\mathbb{E}_t i^{(1)}$	A model or survey	Belief about a future market rate

Policy cuts and rising spot yields

Hypothetical rates	Before	After
Policy target $\bar{i}$	4.00%	3.75%
Current one-year spot $i_t(1)$	4.00%	3.90%
Year-2 expectation + premium	$3.0 + 0.5 = 3.5\%$	$3.4 + 1.0 = 4.4\%$

$$i_t(2): \quad \sqrt{1.040(1.035)} - 1 \longrightarrow \sqrt{1.039(1.044)} - 1.$$

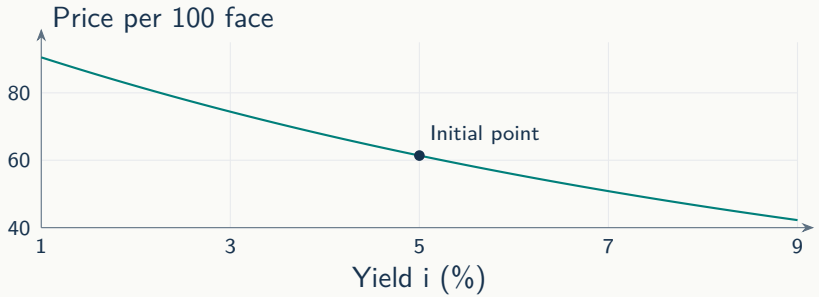
Two-year spot yield: 3.7497% to 4.1497%. The policy target and one-year spot are different rates.

LECTURE 3

Duration and convexity

How does a discount-rate move change the price?

Price sensitivity to a yield change



Fix cash flows and remaining maturity; vary only the yield. That yield comes from the term structure.

Modified duration is the normalized slope

$$D_{\text{mod}} = -\frac{1}{p}\frac{dp}{di}.$$

$$\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta i.$$

A 50 bp increase means $\Delta i = 0.005$. The negative sign gives the price loss.

A zero's duration follows by differentiation

$$p = F(1+i)^{-T}, \qquad \frac{dp}{di} = -TF(1+i)^{-T-1}.$$

$$\boxed{D_{\text{mod}} = \frac{T}{1+i}.}$$

$$T = 10, \quad i = 5\% \quad \Rightarrow \quad D_{\text{mod}} = 9.5238.$$

Coupon duration weights the payment dates

Date k	Cash flow	PV weight
1	4	3.92%
2	4	3.73%
3	104	92.35%

$$D_{\text{Mac}} = \sum_k k\, w_k = 2.8844 \text{ years.}$$

Macaulay and modified duration

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+i}.$$

$$D_{mMac} = 2.8844 \quad \Rightarrow \quad D_{\text{mod}} = 2.7470 \quad \text{at } i = 5\%.$$

Annual coupon convention. Payment timing gives Macaulay duration; the derivative gives modified duration.

Dollar value of one basis point: DV01

DV01 means the dollar value of a one-basis-point yield move.

$$DV01 \approx V D_{\mathrm{mod}} \times 10^{-4}.$$

$$V = \$1{,}000{,}000, \quad D_{\mathrm{mod}} = 8 \quad \Rightarrow \quad DV01 = \$800/\mathrm{bp}.$$

One bp $= 0.01$ percentage point $= 0.0001$. A 1 bp rise costs a long position about \$800.

A ten-basis-point move

$$\Delta i = +10 \text{ bp} \quad \Rightarrow \quad \Delta V \approx -800 \times 10 = -8{,}000.$$
$$\Delta i = -10 \text{ bp} \quad \Rightarrow \quad \Delta V \approx +8{,}000.$$

The linear approximation is symmetric. The exact price curve is curved.

Duration comes from a Taylor expansion

$$p(i + \Delta i) = p(i) + p'(i)\Delta i + \frac{1}{2}p''(i)(\Delta i)^2 + O((\Delta i)^3).$$

The first derivative is the tangent. The second derivative measures curvature.

Cash flows and maturity are fixed while the yield parameter changes.

Convexity normalizes the second derivative

$$\mathcal{C} = \frac{1}{p}\frac{d^2 p}{di^2}.$$

$$\boxed{\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta i + \frac{1}{2}\mathcal{C}(\Delta i)^2.}$$

For a noncallable bond with fixed positive payments, the price–yield convexity is positive.

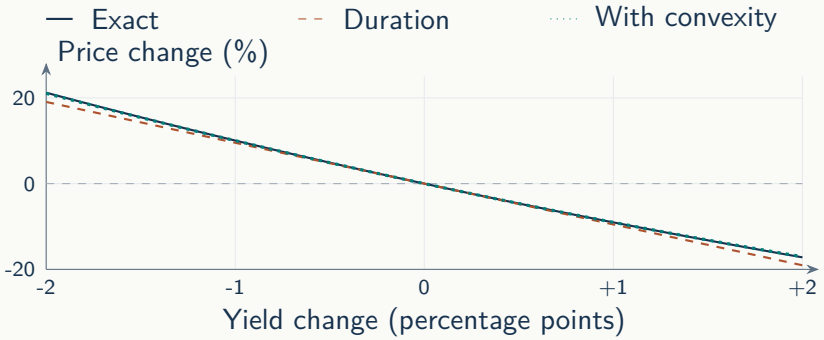
A zero's convexity is explicit

$$\frac{d^2p}{di^2} = T(T+1)F(1+i)^{-T-2}.$$

$$\boxed{\mathcal{C} = \frac{T(T+1)}{(1+i)^2}.}$$

$$T = 10, \quad i = 5\% \quad \Rightarrow \quad \mathcal{C} = 99.7732.$$

Exact, tangent and quadratic repricing



Ten-year zero, initial yield 5%, face 100. The quadratic follows the exact curve more closely locally.

Work out a one-percentage-point move

Yield move	Exact	Duration	With convexity
-1 point	+10.042%	+9.524%	+10.023%
+1 point	-9.043%	-9.524%	-9.025%

$$\tfrac{1}{2}\mathcal{C}(0.01)^2 = 0.0049887 \approx 0.499\%.$$

The asymmetry has economic meaning

For equal rate moves, the gain from a fall exceeds the loss from a rise.

More distant fixed payments generally amplify the curvature.

Positive convexity is valuable, but securities with different convexity need not have the same price or yield.

Sensitivity is conditional on the experiment

A single-yield bump isolates one price-risk channel.

Curve twists, credit spreads and changing cash flows need additional exposures.

Duration matching removes a local parallel-move exposure. It does not make a portfolio riskless.

A liability changes what counts as risky

Ten-year nominal liability of 100	Investment choice
Match the cash flow	A ten-year default-free zero pays 100 on the required date
Roll short bills	The future reinvestment path is uncertain
Trade a long bond over one year	The sale price is uncertain

A bond fund usually keeps exposure beyond one maturity

A single bond has a contractual maturity payment.

A rolling bond fund continually replaces holdings and retains a duration exposure.

The relevant comparison is the investor's cash-flow need and holding horizon.

LECTURE 3

Hedging and market cases

How does a finance view become a position with a measurable risk?

News lab: income horizon or price trade?

A WSJ bond-strategy column asks investors to identify their objective.

Compare a fixed spending need with a forecast of a bond-price rally.

Telis Demos, WSJ, 8 October 2026. Discussion prompt, not a trade recommendation.

Read the WSJ column

News lab: buying during a Treasury sell-off

FT reporting describes investors adding Treasuries after a substantial rise in yields.

A higher starting yield improves the quoted entry rate; a further rate rise can still create a near-term loss.

Clarfelt and Duguid, Financial Times, 13 October 2026. Separate the report from our hypothetical exercise.

Read the Financial Times article

Two real auctions, two different questions

Auction	Offering	High yield	Bid-to-cover
7 Oct: ten-year note reopening	39 bn	5.300%	2.77
8 Oct: thirty-year bond reopening	22 bn	5.618%	2.54

TreasuryDirect results, October 2026. These are auction yields, not coupon rates or total returns.

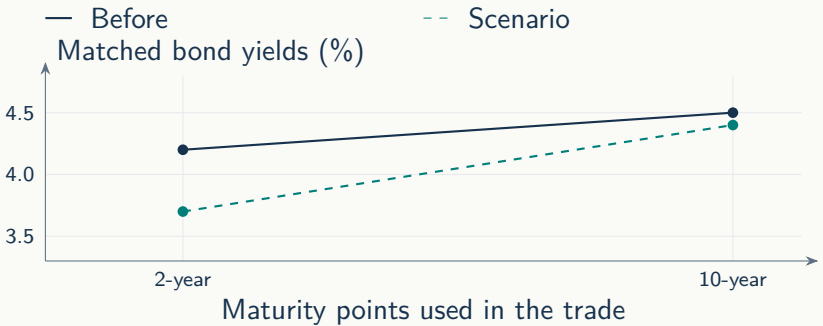
An auction tail needs a market benchmark

$$\text{Tail} = \text{auction stop yield} - \text{when-issued yield at the auction deadline.}$$

$$5.30\% - 5.28\% = 0.02 \text{ points} = 2 \text{ bp.}$$

Illustrative numbers. Same issue, timestamp and quote convention are required.

A curve view needs two maturity scenarios



Two risk points in an invented trade scenario. This is not a sparse empirical curve estimate.

This scenario is a bull steepening

$$\text{Slope} = i_{10} - i_2.$$

$$4.50\% - 4.20\% = 0.30\% \quad \longrightarrow \quad 4.40\% - 3.70\% = 0.70\%.$$

Yields fall. The long-minus-short yield spread widens.

Expressing the relative view

Leg	Benefit	Risk in this scenario
Buy two-year bond	Its yield falls 50 bp	Positive price contribution
Short ten-year bond	Its yield rises	Its 10 bp fall produces a loss

A slope view needs a risk hedge. Equal face amounts do not match rate sensitivity.

Match DV01, rather than market values

Position	Market value	Modified duration	DV01
Long two-year	1,000,000	1.8	180 / bp
Short ten-year	?	8.0	180 / bp

$$V_{\text{short}} = \frac{180}{8(0.0001)} = 225{,}000.$$

Work out the scenario profit and loss

Leg	Yield move	Approximate P&L
Long two-year	-50 bp	$+180(50) = +9{,}000$
Short ten-year	-10 bp	$-180(10) = -1{,}800$
Combined	40 bp steepening	$+7{,}200$

First-order repricing before carry, financing, trading costs and convexity.

A small parallel move cancels at first order

$$\Delta i_2 = \Delta i_{10} = +25 \text{ bp.}$$

$$\Delta V \approx -180(25) + 180(25) = 0.$$

The spread does not change. Higher-order risk and financing remain.

Counterfactual: the curve flattens

Leg	Alternative yield move	Approximate P&L
Long two-year	-10 bp	$+1{,}800$
Short ten-year	-50 bp	$-9{,}000$
Combined	40 bp flattening	$-7{,}200$

Both bonds rally, but this steepener loses.

Price impact of the policy-cut scenario

$$\Delta i_2 \approx +40 \text{ bp} = +0.004.$$

$$V = 1{,}000{,}000, \quad D_{\text{mod}} = 1.8 \quad \Rightarrow \quad \Delta V \approx -7{,}200.$$

$$\mathcal{C} = 5 \quad \Rightarrow \quad \tfrac{1}{2} V \mathcal{C} (0.004)^2 = +40.$$

Approximate repricing: $-7{,}160$. The yield shock is rounded; carry and financing are excluded.

Questions before taking the trade

Which future cash flow, rate shock and holding horizon are being priced?

How does the forecast differ from the forward curve or market baseline?

What happens under the opposite curve move, after financing and costs?

LECTURE 3

Optional foundations

How do dates, states and compounding fit one pricing framework?

General forward intervals

$$[1 + f_t(T_1, T_2)]^{T_2 - T_1} = \frac{d_t(T_1)}{d_t(T_2)}.$$

$$f_t(T_1, T_2) = \left[\frac{d_t(T_1)}{d_t(T_2)}\right]^{1/(T_2 - T_1)} - 1.$$

Annual-effective forward convention. Actual contracts can use simple interest and specific day counts.

Continuous and annual-effective spot rates

$$d_t(T) = e^{-z_t(T)T} = [1 + i_t(T)]^{-T}.$$

$$i_t(T) = e^{z_t(T)} - 1.$$

The Fed and ECB zero inputs use continuous rates z. Our matched-date comparison converts them to i.

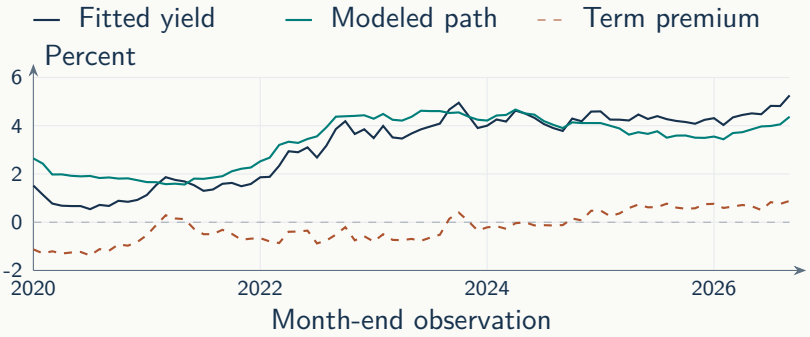
The par coupon comes from discount factors

$$F = c_{\mathrm{par}} F \sum_{k=1}^{n} d_t(k) + F d_t(n).$$

$$\boxed{c_{\mathrm{par}} = \frac{1 - d_t(n)}{\sum_{k=1}^{n} d_t(k)}.}$$

Annual coupon dates and unit year fractions in this simplified formula.

Optional: ACM estimates



NY Fed ACM model, ten-year maturity. January 2020–September 2026. Components are estimates.

ACM components use a model convention

$$4.37\% + 0.89\% = 5.26\% \quad \text{approximately.}$$

30 September 2026	ACM estimate
Modeled rate-path component	4.3740%
Term premium	0.8850%
Fitted ten-year yield	5.2590%

An additive reporting convention of this model. The core pricing relation compounds forwards.

Two states: equal average payoff, different price

Payoff	Good state	Bad state	Expected payoff	Price
A	100	0	50	40.00
B	0	100	50	56.15

Original two-state example. Both states have probability 1/2. Payments in the bad state are more valuable.

A stochastic discount factor prices states

$$p_t = \mathbb{E}_t[m_{t+1} X_{t+1}].$$

$$p_t = \mathbb{E}_t[m]\,\mathbb{E}_t[X] + \mathrm{Cov}_t(m, X).$$

X: future payoff. m: value today per unit paid in that state, adjusted for its probability.

Optional liability hedge: match value and duration

Liability: 100 in ten years, $i = 5\%$.

$$w_1 + w_{20} = 1, \qquad 1w_1 + 20w_{20} = 10.$$

$$w_1 = \frac{10}{19}, \qquad w_{20} = \frac{9}{19}.$$

Weights are current market-value shares in one- and twenty-year zeros.

LECTURE 3

Reading

Where can we deepen the finance and market perspective?

Reading and next questions

Bodie, Kane and Marcus: pricing, term structure and bond-risk management.

The two supplied finance courses: replication, realized returns, auctions and immunization.

Fed, Treasury and ECB sources; WSJ and FT discussion readings via MIT.

Exact chapters, page mappings, dates and links are in the speaker notes and reading list.

Foreign exchange

How do returns, expectations and purchasing power price a currency?

Today's route

Fix the quote and compare covered and uncovered returns.

Use purchasing power as a long-run benchmark.

Combine sticky prices and asset-market equilibrium to explain overshooting.

LECTURE 4

Currency quotes and contracts

What exactly are you buying when a currency strengthens?

One quote convention throughout

E_t = home currency units per one foreign unit.

E rises: the home currency depreciates.

E falls: the home currency appreciates.

A quotation example

Quote	Before	After
JPY per USD	150	135
JPY cost of USD 1,000	150,000	135,000

The yen appreciates. One dollar buys fewer yen.

Spot, forward and FX swap

Contract	Currency exchange dates
Spot	Near settlement at today's agreed rate
Forward	One future exchange at a rate agreed today
FX swap	An exchange now and a reverse exchange later

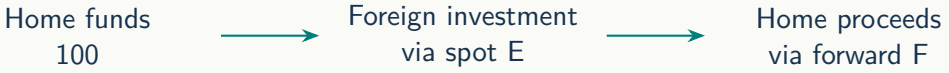
A forward FX rate is a contract price, not a promise about the later spot rate.

LECTURE 4

Covered interest parity

Can two locked currency routes pay different amounts?

Two routes for the same home-currency payoff



Compare this hedged route with lending directly at home.

The locked payoff equation

$$100(1 + R^{\text{home}}) = \frac{100}{E}(1 + R^{\text{foreign}})F.$$

$$\boxed{\frac{F}{E} = \frac{1 + R^{\text{home}}}{1 + R^{\text{foreign}}}.}$$

Covered interest parity, one matched period and consistent conventions.

R^{home}, R^{foreign}: matched-period rates. E: spot quote. F: forward quote.

A complete covered-parity calculation

$$E = 0.90 \text{ EUR per USD}, \quad R^{\mathrm{EUR}} = 2\%, \quad R^{\mathrm{USD}} = 5\%.$$

$$F = 0.90\frac{1.02}{1.05} = 0.874286 \text{ EUR per USD}.$$

$$100 \text{ EUR} \to 111.111 \text{ USD} \to 116.667 \text{ USD} \to 102 \text{ EUR}.$$

Why covered parity can show a basis

Dealer balance sheets, collateral and funding are costly.

The relevant borrowing rate can differ from the lending rate.

A quoted pricing gap need not be a scalable free profit.

LECTURE 4

Uncovered returns and carry

What changes when the future conversion rate is uncertain?

The unhedged foreign payoff

$$\text{Home-currency gross return} = (1 + R^{\text{foreign}})\frac{E_{t+1}}{E_t}.$$

E_{t+1} is the future spot quote. It is unknown today.

Uncovered interest parity is an equilibrium assumption

$$R^{\text{home}} \approx R^{\text{foreign}} + \mathbb{E}_t \Delta \ln E_{t+1}.$$

Equal expected returns require assumptions about risk and beliefs.

Currency risk premium

$$R^{\text{home}} - R^{\text{foreign}} - \mathbb{E}_t \Delta \ln E_{t+1} = \rho_t.$$

ρ_t: expected excess return required on the home asset.

The rate differential mixes expected FX change and compensation for risk.

Carry: borrow yen and invest dollars

$$E_t = 150 \text{ JPY per USD}, \quad R^{\text{JPY}} = 1\%, \quad R^{\text{USD}} = 5\%.$$

Borrow JPY 150 and buy USD 1.

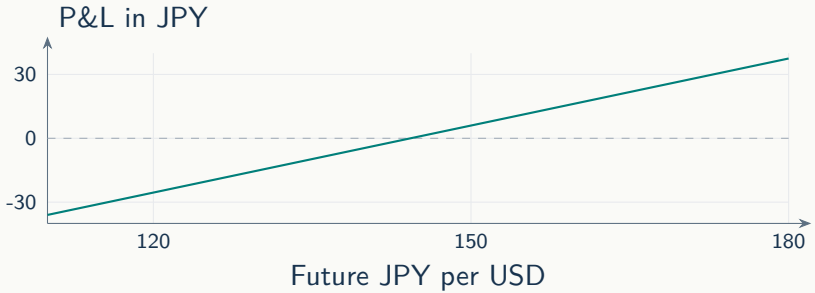
Next year receive USD 1.05 and owe JPY 151.50.

A small interest gain can meet a large FX loss

Future JPY per USD	JPY proceeds	Debt due	P&L
150	157.50	151.50	+6.00
135	141.75	151.50	-9.75

In the second scenario, JPY per USD falls 10%. Figures are hypothetical.

The carry payoff depends on the exit exchange rate



Break-even: 151.50 / 1.05 = 144.29 JPY per USD, before costs.

Crash risk can hide behind frequent small gains

Toy outcome	Probability	Normalised P&L
Quiet period	95%	+4%
Funding-currency surge	5%	-60%

$$0.95(4\%) + 0.05(-60\%) = 0.8\%.$$

An invented distribution, not an estimate of any currency strategy.

LECTURE 4

Expectations and purchasing power

Why can a rate increase mean appreciation now or weakness over time?

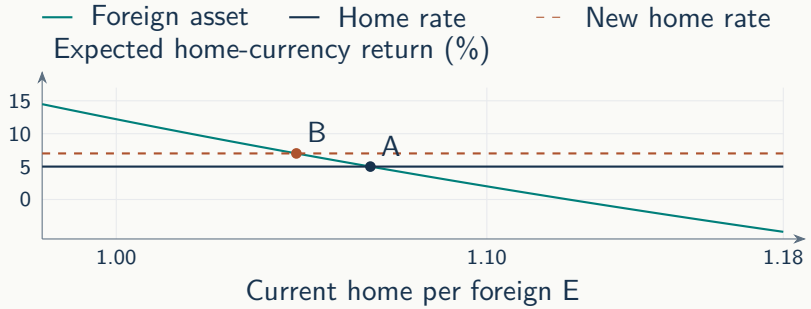
Short-run FX as an asset price

$$E_t = \frac{(1 + R^{\text{foreign}})E_t^e}{1 + R^{\text{home}}}.$$

E_t^e: expected next-period quote, fixed for this experiment.

A higher home rate then supports a stronger home currency.

A short-run FX return diagram



Expected future E = 1.10 and foreign rate 2%, both fixed. Hypothetical values.

The rate and expectation experiments

Change, other inputs fixed	Current home-per-foreign E
Home interest rate rises	Falls: home appreciates
Expected future E rises	Rises: home depreciates
Required home risk premium rises	Rises: home weakens

The third result uses the premium convention defined earlier.

Purchasing power parity

$$E^{PPP} = \frac{Q^{\text{home}}}{Q^{\text{foreign}}}.$$

A comparable basket costs 100 at home and 80 abroad.

PPP implies 1.25 home units per foreign unit.

Why a cheap haircut is hard to arbitrage

Haircuts require local labour and premises.

National baskets have different weights and taxes.

Productivity can change equilibrium relative prices.

Productivity and local service prices

Toy economy	Tradable productivity	Wage	Haircut cost
A	10 units / hour	10 / hour	10
B	5 units / hour	5 / hour	5

Tradable price 1 in both economies. One haircut per hour. Mobile domestic labour.

The real exchange rate

$$q = \frac{EQ^{\text{foreign}}}{Q^{\text{home}}}.$$

q: foreign-basket price relative to the home basket.

A rise in q is a real depreciation of the home currency.

Relative PPP and the Fisher effect

$$\Delta \ln E \approx \pi^{\text{home}} - \pi^{\text{foreign}}.$$

Persistent higher home inflation tends to erode the home currency.

Its higher nominal rate may simply compensate for that inflation.

Real interest differences and real FX expectations

$$R^{\text{real,home}} - R^{\text{real,foreign}} \approx \mathbb{E}_t \Delta \ln q + \rho_t.$$

Equal real rates need additional assumptions.

Expected real FX changes and risk premia can support persistent differences.

LECTURE 4

Dornbusch overshooting

Why can the currency jump further than its long-run adjustment?

A permanent monetary level shock

The money stock rises permanently by 10% in the classic model.

Prices are sticky initially. Long-run output is fixed.

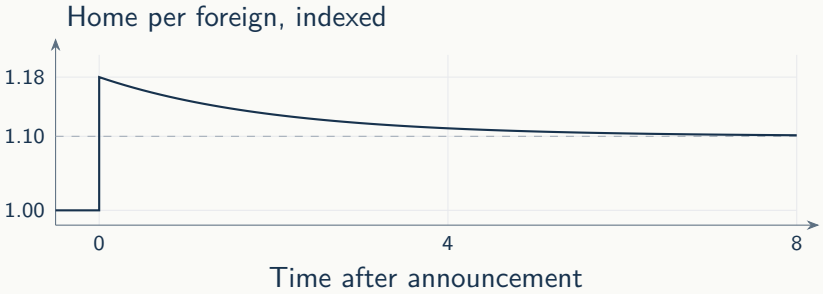
Flexible asset prices respond immediately.

Prices adjust slowly



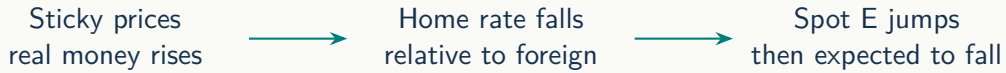
Original illustrative transition. Initial level 1, long-run level 1.10.

The exchange rate can overshoot



The home currency weakens sharply, then appreciates toward its new long-run level.

The overshooting mechanism



An expected future appreciation reconciles the lower home interest rate.

Exercise: three explanations for a high yield

Case A: an unexpected tightening, fixed future inflation and FX expectations.

Case B: higher expected persistent inflation.

Case C: a larger currency risk premium.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.

Open-economy policy and regimes

How does currency adjustment interact with demand and policy constraints?

Today's route

Trade prices and quantities determine the spending effect of FX.

DD–AA connects output with asset-market equilibrium.

Pegs, floating rates and the trilemma change the policy experiment.

LECTURE 5

Trade prices and quantities

Does a depreciation automatically improve the trade balance?

Exporters and importers face different cash flows



Invented trading economy. Imported inputs can be essential to domestic production.

The real exchange rate and net exports

$$q = \frac{EQ^{\text{foreign}}}{Q^{\text{home}}}, \qquad NX = NX(q, Y, Y^{\text{foreign}}).$$

A higher q makes home goods relatively cheaper.

Higher home income can raise imports.

A depreciation raises the local import bill first

Balanced trade initially	Before	Immediate effect
Real exchange rate q	1.00	1.10
Exports, fixed quantity and price	100	100
Import quantity, fixed	100	100
Home value of imports	100	110
Net exports	0	-10

Producer-currency prices are fixed. Quantities have not adjusted.

Quantities can later reverse the result

After the same depreciation	Value
Export volume rises 8%	108
Import volume falls 6%	94
Import value: 1.10×94	103.40
Net exports: 108 - 103.40	+4.60

Invented quantity responses, after enough time to adjust.

Marshall–Lerner gives a local condition

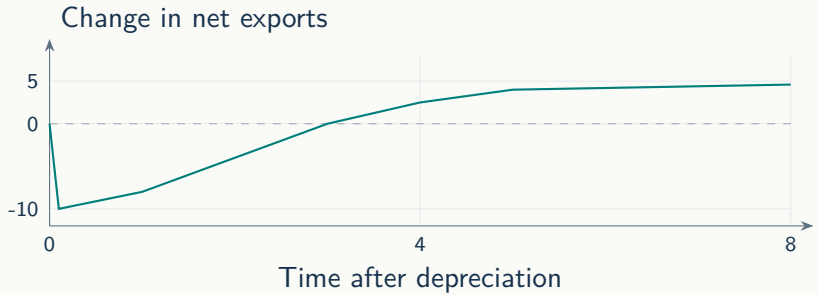
$$\epsilon_X + \epsilon_M > 1.$$

ϵ_X: export-volume elasticity to real depreciation.

ϵ_M: absolute import-volume elasticity.

Balanced starting trade and fixed producer prices in the simplest derivation.

The J-curve is a timing mechanism



Original hypothetical path: the import price effect precedes quantity adjustment.

Invoice currency changes pass-through

Euro exporter to a dollar buyer	Initial	After euro depreciation
USD per EUR	1.10	1.00
EUR invoice: EUR 100 in USD	110	100
USD invoice: USD 110 in USD	110	110

Same exchange-rate move. Different invoice contracts. Hypothetical prices.

The dollar can matter in third-country trade

A UK importer buys from Korea at an invoice price of USD 100.

GBP per USD rises from 0.80 to 0.88.

The sterling bill rises from GBP 80 to GBP 88.

LECTURE 5

The DD schedule

Which output and exchange-rate pairs balance the goods market?

A short-run open-economy model

$$Y = C(Y - T) + I + G + NX(q, Y, Y^{\text{foreign}}).$$

Goods prices, foreign conditions and expected future FX are fixed.

For this experiment, investment I is fixed too.

A numerical demand equation

$$D = 40 + 0.5Y + 10E.$$

D: planned spending on home output.

A higher E raises demand at each output level.

Only half of extra output returns as extra planned spending.

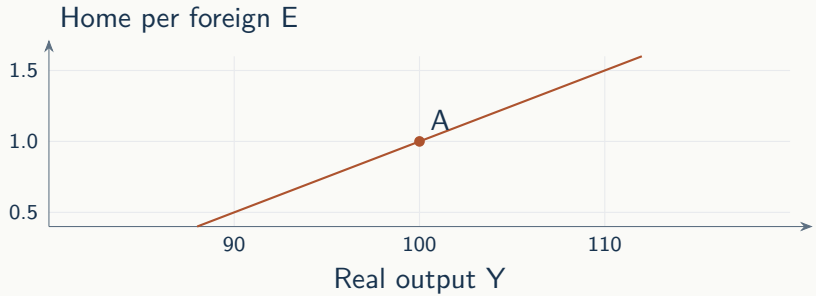
Goods-market equilibrium gives DD

$$Y = D \quad \Rightarrow \quad Y = 80 + 20E.$$

A weaker currency raises demand and equilibrium output.

DD slopes upward with Y horizontal and E vertical.

DD as a graph



Original numerical goods-market equilibrium: $Y = 80 + 20E$.

LECTURE 5

The AA schedule

What second relationship pins down output and the currency?

Money demand and FX pricing together

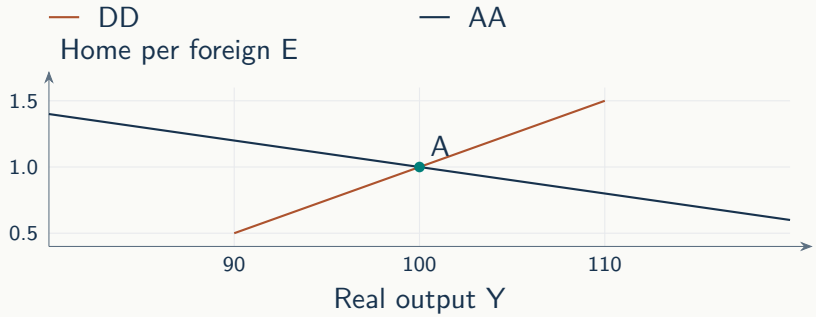
$$M/Q = L(Y, R^{\text{home}}).$$

$$R^{\text{home}} \approx R^{\text{foreign}} + \mathbb{E}_t \Delta \ln E.$$

Higher Y raises money demand and the home rate.

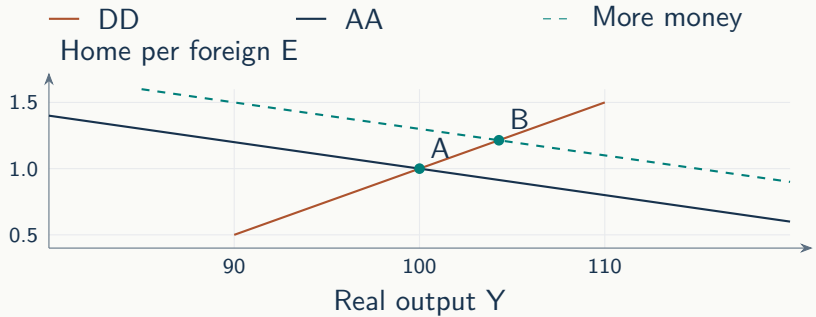
At fixed expected future FX, that supports home appreciation.

AA slopes downward



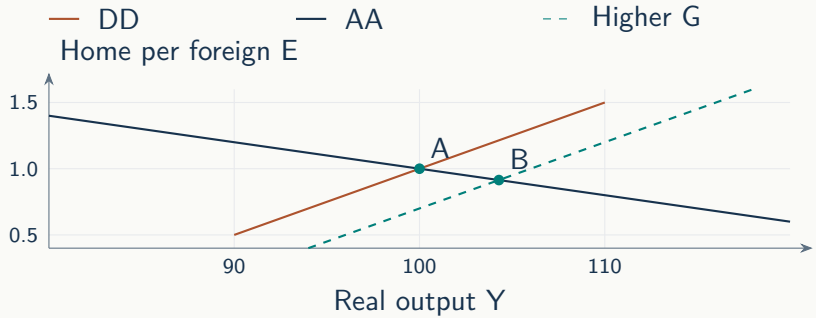
The intersection A balances goods and asset markets. Original linear illustration.

Temporary monetary expansion under a float



AA shifts up. In this model output rises and the home currency depreciates.

Temporary fiscal expansion under a float



DD shifts right. Output rises, while currency appreciation offsets some demand.

A fiscal expansion need not improve net exports

Government demand rises directly.

Higher income can increase imports.

Appreciation can reduce expenditure switching toward home goods.

At the lower bound, expectations still matter

Cash limits how far nominal rates can be pushed below zero.

Extra reserves may have little effect on the current short rate.

Credible changes in future policy can still affect real rates and FX.

LECTURE 5

Pegs and interventions

What changes when the central bank promises a fixed currency price?

The promise behind a peg

The central bank stands ready to exchange currencies at the peg.

It must supply foreign reserves when investors want to exit.

The promise constrains monetary policy when capital moves freely.

Defending a peg changes the balance sheet

Central bank transaction	Assets	Liabilities
Sell 10 of foreign reserves	Foreign assets -10	
Receive home reserve balances		Bank reserves -10

An unsterilised sale drains domestic liquidity.

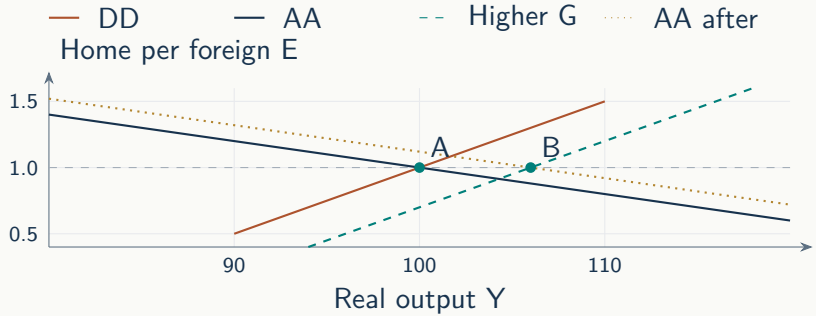
Sterilisation offsets the liquidity effect

Second transaction	Assets	Liabilities
Buy 10 of domestic bonds	Domestic assets +10	Bank reserves +10

Total monetary liabilities return to their initial size.

The asset portfolio now contains fewer foreign and more domestic assets.

A fixed peg removes one degree of freedom



At E = 1, higher G raises output from 100 to 106 if monetary policy accommodates.

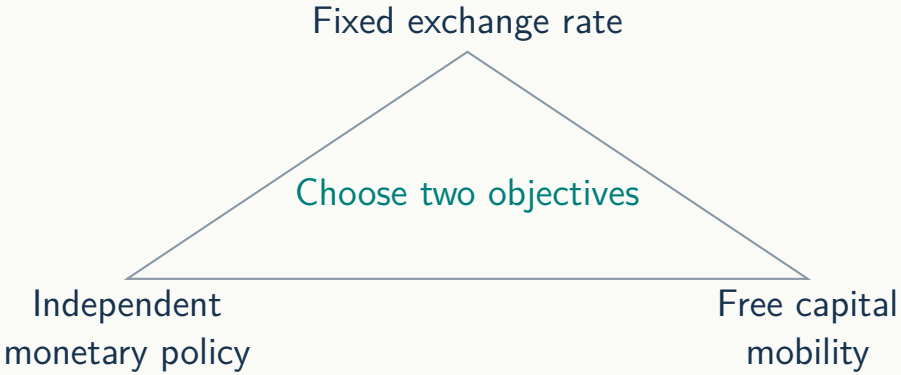
Independent monetary expansion conflicts with the peg

A rate cut creates pressure for home-currency depreciation.

Defending the peg withdraws liquidity or reverses the rate cut.

With an unchanged credible peg, the independent impulse cannot persist.

The monetary trilemma



LECTURE 5

Gold and the international system

What does a monetary anchor buy, and what adjustment does it impose?

Gold convertibility



Illustration of redemption, not a historical photograph or certificate.

Gold fixes a currency relationship

$$1 \text{ home unit} = a \text{ grams}, \qquad 1 \text{ foreign unit} = b \text{ grams}.$$

$$E = \frac{b}{a} \quad \text{home units per foreign unit}.$$

Transport, insurance and conversion costs create a band around parity.

Deflation makes a fixed nominal rate tighter

$$R^{\text{nom}} = 0\%, \quad \mathbb{E}\pi = -5\% \quad \Rightarrow \quad R^{\text{real}} \approx 5\%.$$

Falling expected prices can raise the real cost of borrowing.

A monetary anchor can constrain the response.

Three monetary arrangements

Arrangement	Anchor	Adjustment problem
Gold standard	Gold convertibility	Domestic demand may bear the strain
Bretton Woods	Dollar pegs and dollar-gold link	Reserve demand versus convertibility
Floating fiat currencies	Domestic policy frameworks	FX absorbs shocks, can also amplify them

Triffin's dilemma

Global demand for dollar reserves requires dollar liabilities.

More claims relative to gold can weaken confidence in convertibility.

Providing liquidity and preserving a fixed gold promise can conflict.

Counterfactual exercise: who can ease?

Economy	Regime	Can cut independently?
A	Float, mobile capital	?
B	Credible peg, mobile capital	?
C	Peg, effective capital controls	?

Assume the foreign policy rate stays fixed and no devaluation is expected.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.

Currency crises and the euro

How can balance sheets and expectations turn a funding shock into a crisis?

Today's route

Currency mismatches can reverse the trade benefits of depreciation.

Pegs can fail through deteriorating fundamentals or coordination.

A currency union changes the adjustment tools and financial risks.

LECTURE 6

Balance-sheet exposure

Why can a weaker currency reduce domestic demand?

A firm borrows dollars but earns home currency

Balance sheet, home currency	Before	After depreciation
Operating assets	200	200
USD debt: one dollar	150	180
Equity	50	20

Home per USD rises 20%, from 150 to 180. Asset values are held fixed.

Trade and financial channels can oppose each other



A currency move changes liabilities as well as product prices.

Debt and equity share bad states differently

Foreign finance	What happens in a recession?
Fixed nominal debt	Promised payment initially stays fixed
Equity ownership	Dividends and market value can absorb losses

Equity shares risk, while financing terms, control and valuation still matter.

LECTURE 6

Pegs under pressure

When does defending a fixed rate become impossible or too costly?

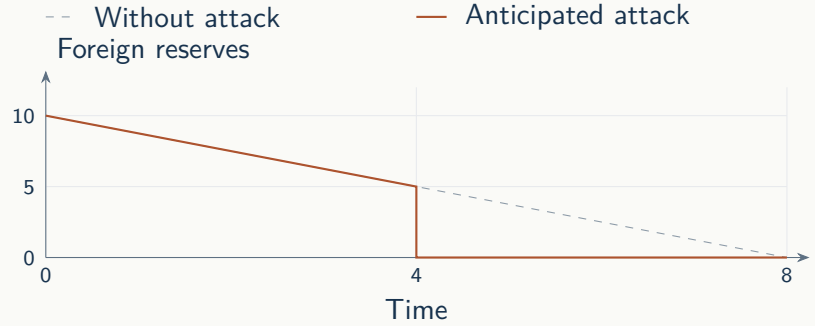
A first-generation crisis mechanism

Domestic credit keeps expanding while the peg fixes money demand.

Foreign reserves offset the expansion and decline.

The policy combination eventually cannot continue.

Reserves can be exhausted by an anticipated attack



Schematic path, not a fitted crisis date or empirical reserve series.

A second-generation crisis mechanism

Defending the peg raises interest rates and domestic economic costs.

Investors' exit expectations can increase the cost of defence.

The authorities may choose to abandon the peg.

A coordination example

Two large investors can each withdraw 4 units.

The authority can absorb 6 units of withdrawals.

One exit is manageable. Two simultaneous exits break the peg.

The coordination game

Investor 1 / Investor 2	Stay	Exit
Stay	(0, 0)	(0, -1)
Exit	(-1, 0)	(+2, +2)

Both staying and both exiting can be equilibria.

A backstop can change incentives

Credible support can remove the payoff from joining a run.

Support needs resources, legal authority and believable conditions.

A liquidity backstop does not automatically repair insolvency.

LECTURE 6

Currency unions

When is sharing one monetary policy worth the loss of local adjustment?

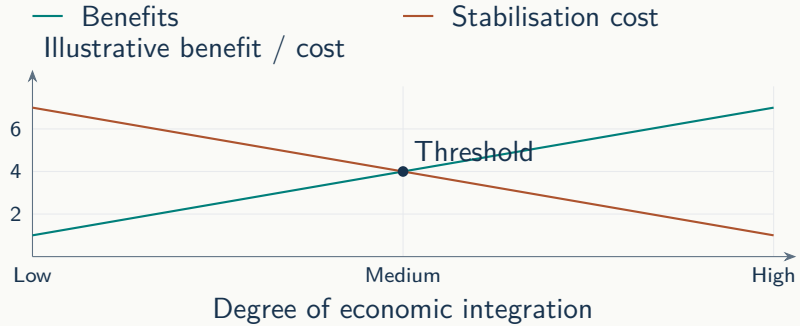
The benefits and costs of a common currency

Trade and financial integration reduce conversion costs and currency risk.

A common rate cannot respond separately to every national shock.

Other adjustment channels become more important.

An integration trade-off



A schematic OCA trade-off, not a measured threshold for any nation.

A common currency does not make every price identical

Euro-area countries share monetary policy.

Their consumption baskets, energy exposure, local taxes and price adjustments still differ.

One currency does not imply one national inflation rate.

Different baskets can produce different inflation

Hypothetical basket	Country A	Country B
Energy weight	10%	30%
Other goods/services weight	90%	70%
Energy price increase	20%	20%
Other price increase	2%	2%
Basket inflation	3.8%	7.4%

Same component price increases, different fixed basket weights.

A shared nominal rate can mean different real rates

Hypothetical comparison	Country A	Country B
Same nominal rate	4%	4%
Expected basket inflation	2%	4%
Approximate expected real rate	2%	0%

The purchasing-power cost depends on expected inflation.

Euro government bonds can also have different yields

Same currency, different issuers and credit risk.

Liquidity, collateral value and market conditions also differ.

National inflation is not a mechanical add-on to a euro sovereign yield.
The plotted ECB AAA curve is a benchmark sample, not every nation's curve.

Relative prices can adjust without a currency change

$$\Delta \ln q = \pi^{\text{foreign}} - \pi^{\text{home}} \quad \text{when } E \text{ is fixed.}$$

Higher local inflation can create real appreciation inside a currency union.

LECTURE 6

Banks and sovereigns

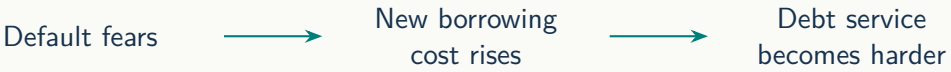
How can two balance sheets make each other weaker?

Banks transform maturities

Assets	Liabilities
Longer-term loans and securities	Deposits withdrawable on demand

A solvent bank can lack cash for simultaneous withdrawals.

A sovereign refinancing feedback



The higher funding cost can make the original fear more plausible.

The bank–sovereign loop



A historical backstop: OMT in 2012

The ECB set out its OMT technical framework on 6 September 2012.

Eligible sovereign support required programme conditionality.

The mechanism targeted severe transmission and redenomination fears.

Exercise: liquidity support or a fiscal repair?

A bank has sound long-term loans but sudden deposit withdrawals.

A government faces a temporary market closure with credible future surpluses.

Another government cannot plausibly fund its promises in real terms.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.

Public debt and macro trades

How do fiscal promises, monetary policy and market prices interact?

Today's route

Derive the debt ratio before debating debt sustainability.

Distinguish monetary control from fiscal and rollover constraints.

Use dated US/Japan evidence to build conditional trading scenarios.

LECTURE 7

Debt arithmetic

What makes the debt ratio rise or fall?

The government's one-period financing constraint

$$B_t = (1 + R_t^{\text{debt}})B_{t-1} - PB_t.$$

Symbol	Meaning
B_t	Nominal debt outstanding after this period
R_t^{debt}	Effective nominal interest rate on existing debt
PB_t	Primary surplus: revenues less non-interest spending

Dividing by nominal GDP

$$d_t \equiv \frac{B_t}{GDP_t^{\text{nominal}}}, \qquad pb_t \equiv \frac{PB_t}{GDP_t^{\text{nominal}}}.$$

$$\boxed{d_t = \frac{1 + R_t^{\text{debt}}}{1 + g_t^{\text{nominal}}} d_{t-1} - pb_t.}$$

g_t^{nominal}: nominal GDP growth over the same period.

The interest–growth snowball

$$\Delta d_t = \frac{R_t^{\text{debt}} - g_t^{\text{nominal}}}{1 + g_t^{\text{nominal}}} d_{t-1} - pb_t.$$

Interest above growth pushes the ratio up.

A primary surplus pushes the ratio down.

A complete debt-ratio calculation

$$d_{t-1} = 100\%, \quad R^{\text{debt}} = 6\%, \quad g^{\text{nominal}} = 3\%, \quad pb = 1\%.$$

$$\Delta d = \frac{0.06 - 0.03}{1.03} \times 1 - 0.01 = 1.91\% \text{ of GDP.}$$

The debt ratio rises from 100% to about 101.91%.

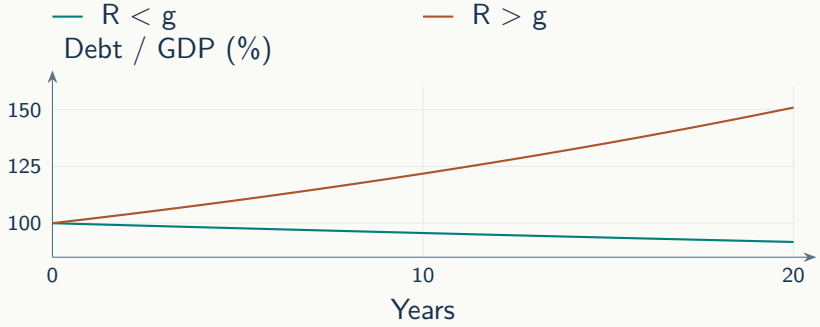
Growth above interest can permit a primary deficit

$$R^{\text{debt}} = 4\%, \quad g^{\text{nominal}} = 5\%, \quad d_{t-1} = 100\%.$$

$$pb^{\text{stabilising}} = \frac{0.04 - 0.05}{1.05} \times 1 = -0.95\% \text{ of GDP.}$$

A conditional arithmetic result. Financing risk and future rates remain.

Small parameter differences compound



Invented constant paths: green pb = -0.5%, red pb = +1% of GDP.

A rising ratio is a warning, not an instant default test

Fiscal adjustment can take time.

Risk depends on the distribution of future funding and primary balances.

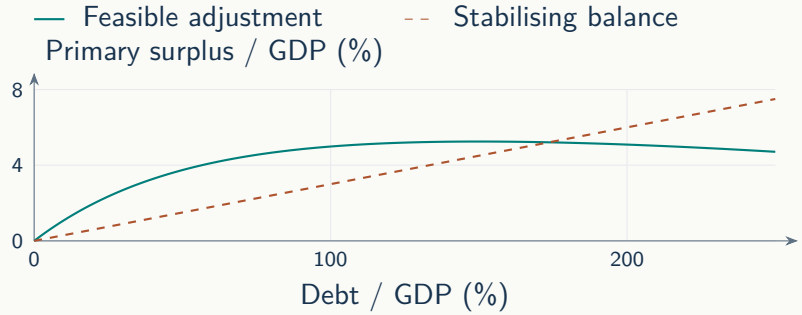
Debt definitions and the holder/currency/maturity structure matter.

A long maturity delays the cash-flow effect

Illustrative refinancing need	Share resetting this year	Rate rise
Mostly short debt	80%	+1 point
Mostly long fixed-rate debt	10%	+1 point

The same market-rate move can have very different near-term budget effects.

Fiscal fatigue and a possible adjustment limit



Invented fiscal-capacity curve and approximate interest–growth gap of 3 points.

LECTURE 7

Fiscal and monetary interaction

Can issuing money remove the government's real resource constraint?

Nominal repayment and real value are different promises

A government with its own currency can create nominal settlement balances.

It cannot create unlimited real goods by changing nominal units.

Inflation, legal constraints and political choices still affect outcomes.

Fiscal dominance

Debt-financing needs constrain the monetary response to inflation.

A central bank may face pressure to keep financing costs low.

That pressure alone does not prove which policy regime actually prevails.

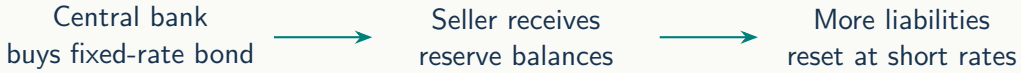
Inflation can reduce some real debt burdens

Unexpected inflation erodes the real value of fixed nominal claims.

Indexed debt and new borrowing costs can offset the benefit.

Tax/spending indexation and credibility change the fiscal result.

QE changes the public sector's liability mix



Consolidated exposure depends on reserve remuneration and the full balance sheet.

Fiscal feedback can stabilise debt in a toy model

$$pb_t = a + b\, d_{t-1}.$$

a: baseline primary balance. b: response to inherited debt.

$$d_t = \left[\frac{1 + R^{\text{debt}}}{1 + g^{\text{nominal}}} - b\right] d_{t-1} - a.$$

LECTURE 7

Dated US and Japan cases

What is verified evidence, and what is a trading hypothesis?

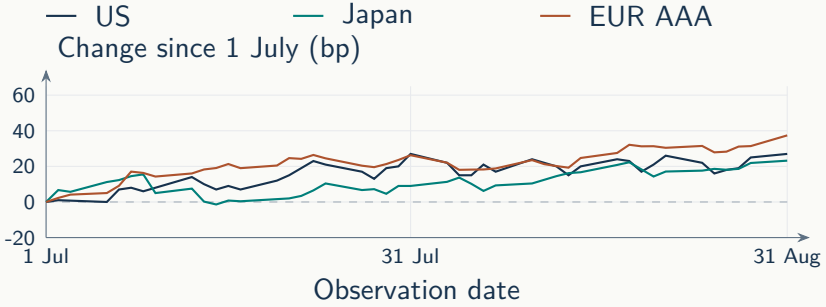
July–August 2026: a common date window

Compare 1 July with 31 August, then inspect the path between them.

Use US Treasury, Japan MOF and ECB daily series.

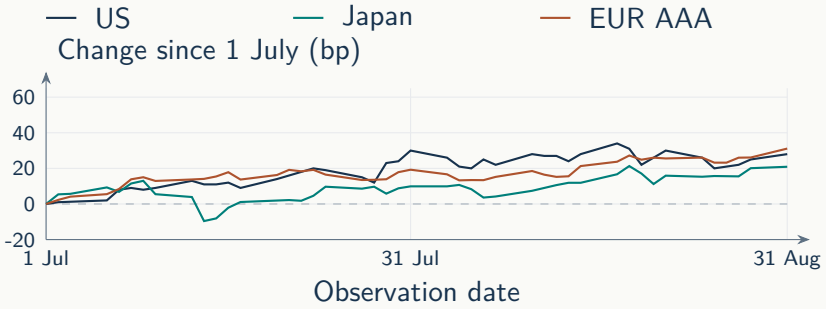
Keep currencies, yield definitions and local observation times explicit.

Summer 2026: ten-year yield changes



Official daily observations, own baseline change. Local closes/fixings differ.

Summer 2026: thirty-year yield changes

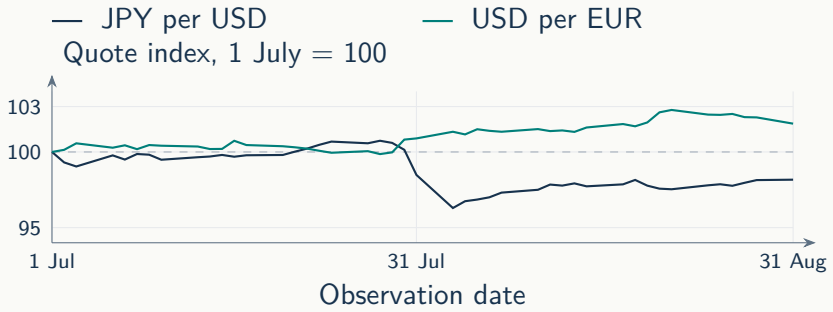


Official daily observations, own baseline change. Local closes/fixings differ.

Summer endpoints: yields rose in all three samples

Series	1 July	31 August	Change
US 10-year	4.480%	4.750%	+27.0 bp
Japan 10-year	2.711%	2.943%	+23.2 bp
EUR AAA 10-year	2.929%	3.303%	+37.4 bp
US 30-year	4.970%	5.250%	+28.0 bp
Japan 30-year	3.883%	4.092%	+20.9 bp
EUR AAA 30-year	3.379%	3.690%	+31.1 bp

Summer FX moved on its own path



Different quote directions: lower JPY per USD strengthens yen; higher USD per EUR strengthens euro.

The cross rate checks the currency story

$$\text{JPY per EUR} = (\text{JPY per USD})(\text{USD per EUR}).$$

ECB reference observation	1 July	31 August
JPY per USD, derived	162.71	159.73
USD per EUR	1.1383	1.1596
JPY per EUR	185.21	185.22

Endpoint comparison. Small net cross-rate change can hide moves within the window.

A rise in yield is not a flow-of-funds explanation

Expected real rates, expected inflation and risk premia can move together.

Hedging costs and relative duration exposure affect investor decisions.

Prices alone do not show which investors repatriated capital.

Exercise: three competing summer explanations

Candidate mechanism	What else would you check?
Higher expected future policy rates	Short-rate contracts and policy surprises
Higher duration / fiscal risk premium	Premium models, supply and risk measures
Cross-border portfolio adjustment	Hedged returns and actual transaction data

Several mechanisms may contribute. The daily charts do not choose one automatically.

US policy evidence: September 2026

On 16 September the Fed raised its target range by 0.25 points to 3.75–4.00%.

The statement described elevated inflation.

The policy decision is one input to pricing the future rate path.

US curve context: 15 October 2026

Observation	Value
Two-year Treasury par yield	4.77%
Ten-year Treasury par yield	5.10%
Ten minus two-year spread	33 basis points

Dated observations. This is a level comparison, not an identified causal decomposition.

Three hypotheses behind a higher US long yield

Hypothesis	Evidence to seek
Higher expected policy path	Dated short-rate pricing and announcements
Higher term/inflation risk premium	Models and inflation-risk indicators
More duration or funding pressure	Issuance, positioning and market functioning

Competing explanations, not an attribution of the observed snapshot.

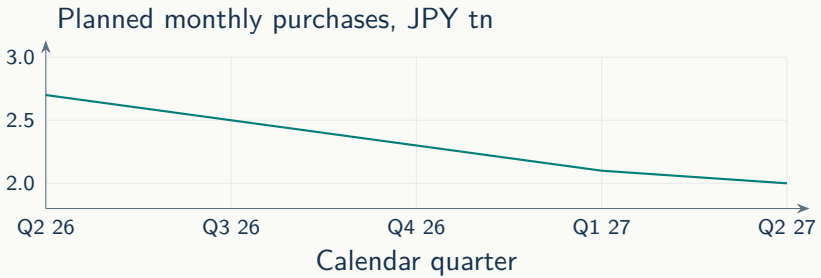
Japan policy evidence: September 2026

The 18 September BoJ decision set the overnight call-rate guideline at around 1.25%.

The new guideline took effect on 24 September.

A higher short rate is one component of normalisation.

Japan: the JGB purchase plan



BoJ plan announced 16 June 2026. Planned amounts, not realised purchases.

Why Japanese capital need not immediately return home

Domestic yields, hedging costs and expected FX returns all matter.

Hedged foreign bonds retain interest-rate and spread exposure.

Mandates, liabilities and existing hedges affect portfolio changes.

A hedge-cost illustration

$$R^{\mathrm{JPY}} = 1\%, \quad R^{\mathrm{USD}} = 4\%, \quad E = 150 \text{ JPY per USD.}$$

$$F = 150\frac{1.01}{1.04} = 145.67 \text{ JPY per USD.}$$

A covered one-year USD deposit returns 1% in yen under parity.

US/Japan counterfactuals

Shock relative to expectations	Conditional channel
BoJ path reprices higher	JPY carry funding becomes less attractive
JGB term premium rises	Long JGB prices fall at fixed cash flows
Global risk appetite collapses	Positions may unwind and premia change

Scenarios for reasoning, not predictions or descriptions of a realised event.

LECTURE 7

A macro trading lab

How does a conditional view become a sized position?

Markets react to the surprise in the full announcement

A 0.25-point cut was expected. A 0.50-point cut arrives.

The decision is 0.25 points more dovish than expected.

Guidance and economic news can change the later path too.

Exercise: a cut today, but a higher long yield?

After the announcement, assume:

Expected average short rate over ten years rises 0.10 points.

The ten-year term premium rises 0.20 points.

What happens to the ten-year yield? What about a duration-8 bond?

The decomposition gives the answer

$$\Delta R^{\text{spot}}(10) \approx 0.10 + 0.20 = 0.30 \text{ percentage points.}$$

$$\frac{\Delta P}{P} \approx -8 \times 0.003 = -2.4\%.$$

The long yield rises and the bond loses value in this specified scenario.

Turn a headline into a conditional chain

Identify the shock and the prior expectation.

State the model and policy response.

Translate the view into dated cash flows and yield changes.

Size the exposure and test the counterfactual.

Final exercise: a trade needs a falsifiable claim

Choose a shock, prior pricing and a policy response.

Predict two yields and one exchange rate, with assumptions.

Size the trade and name the counterfactual that would make it lose.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

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